

Policies change the system around them

Three rigorous economics papers · two current applied-development studies and one field-experiment classic. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

These papers treat policy as a change to an interacting system, not a switch on one outcome. China's straw-burning ban cleans the air but induces farmers to use more chemicals, shifting damage into water. Afghan demining removes a physical barrier and produces reinforcing gains in land use, local activity, and security. Uganda's community-monitoring experiment changes the information and accountability facing health workers, improving effort and child outcomes without adding major resources. The broader literature is increasingly tracing **behavioral responses, spillovers, and complementary institutions**: the net effect of a policy depends on what people substitute toward and whether surrounding systems amplify or dilute the original intervention.

01 CURRENT / GENERAL INTEREST · STARTS ON PAGE 2 · *Journal of Development Economics* 181 · 2026 · peer reviewed · CC BY 4.0

When the Fire Ends: Straw Burning, Regulation, and Pollution Substitution

China's staggered ban on crop-residue burning asks whether a successful environmental rule can merely move pollution elsewhere. A generalized difference-in-differences design combines satellite fire detections, air- and water-quality monitors, and agricultural inputs. The ban sharply reduces fires and particulate pollution, but farmers substitute toward fertilizer and pesticides, worsening water pollution enough to offset much of the estimated health benefit. Event studies, alternative estimators, placebo assignments, and watershed-year controls support the mechanism; remaining uncertainty comes from valuing pollution damages and measuring farm inputs at coarser geographic levels.

02 CURRENT / CONFLICT AND RECONSTRUCTION · STARTS ON PAGE 17 · *Journal of Development Economics* 182 · 2026 · peer reviewed · CC BY 4.0

Landmine Clearance and Economic Development

The paper maps more than 15,000 Afghan hazardous areas cleared since 1992 and uses a post-2008 targeting shift to study what demining unlocks. Staggered difference-in-differences and modern event-study estimators link clearance to brighter nighttime lights, more productive land use, and lower nearby conflict risk. Pre-trend tests and alternative estimators strengthen the causal case, while the authors acknowledge that five-year satellite intervals limit dynamic land-use analysis and that effects may differ where clearance priorities or security conditions are less favorable.

03 CLASSIC / HEALTH AND ACCOUNTABILITY · STARTS ON PAGE 32 · *Quarterly Journal of Economics* 124(2) · 2009 · classic · World Bank open edition

Power to the People: Evidence from a Randomized Field Experiment on Community-Based Monitoring in Uganda

Fifty Ugandan health-facility catchment areas were randomized to receive local report cards and facilitated meetings that helped citizens monitor providers. One year later, treatment communities used clinics more, reported better staff effort, and saw infant weight gains and roughly one-third lower under-five mortality. The randomized design, baseline balance, difference-in-differences checks, and spillover tests are persuasive; the package combines information with organizing support, so the experiment cannot isolate which component caused the improvement or guarantee the same effects under scale-up.

Editorial note: references and appendices were omitted where possible; no included article contributes more than 15 text pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.



Regular Article

When the fire ends: Straw burning, regulation, and pollution substitution

Hai Hong^a, Kevin Chen^{a,b,*}^a China Academy for Rural Development, School of Public Affairs, Zhejiang University, Hangzhou, China^b International Food Policy Research Institute, Beijing, China

ARTICLE INFO

JEL classification:

Q56
Q58
O13
O44
D24

Keywords:

Straw burning
Pollution substitution
Air pollution
Water pollution

ABSTRACT

Environmental regulations can trigger unintended pollution externalities if they lack well-designed economic incentives or fail to account for the responses of polluters. This paper examines the effectiveness and unintended consequences of the Universal Prohibition on Straw Burning (UPSB) policy in China. By exploiting a generalized difference-in-differences design, we find that the UPSB policy significantly reduces agricultural fires and air pollution through top-down campaign-style enforcement. However, as straw burning is commonly used to kill pests and fertilize the soil, the UPSB policy also increases the use of chemical fertilizers and pesticides, leading to magnified water pollution. Cost-benefit analysis suggests that much of the health benefit from improved air quality is offset by the health cost from degraded water quality. Our findings highlight the importance of considering the potential responses of individuals subject to the regulation when conducting policy evaluation.

1. Introduction

Environmental regulations can trigger unintended pollution externalities if they lack well-designed economic incentives or fail to account for the responses of polluters. For instance, when the government controls air pollution with punitive measures, the regulated firms may dissolve harmful gases in liquids, causing regulation-induced water pollution (Gibson, 2019; Greenstone, 2003).¹ Yet, the extent to which such regulations can cause unanticipated negative externalities remains less understood. In this paper, we leverage China's Universal Prohibition on Straw Burning (UPSB) policy as a quasi-experiment, and provide the first evidence of the effectiveness and the unintended pollution externalities of environmental regulation in the agricultural sector.

China has had a long tradition of burning straw since ancient times.² Farmers burn straws for several reasons. First, burning is conducive to

new crops since it can eradicate potential invasive plant species, weeds, fungi, and bacteria (Graff Zivin et al., 2020; Levine, 1991). Second, the ashes after burning can fertilize the soil (He et al., 2020a; Nian, 2023). Lastly, it helps clear the field and save on labor input, preparing the land for the next round of cultivation (Guo, 2021). However, such open-ground burning activities also heavily contribute to severe air pollution, with detrimental effects documented on health and economic outcomes (Graff Zivin et al., 2020; Guo, 2021; He et al., 2020a; Lai et al., 2022; Rangel and Vogl, 2019).³

In response to the deteriorating air quality caused by straw burning, the Chinese government has introduced the UPSB policy across provinces since 2013. In practice, to motivate local officials (mostly county and prefectural leaders), provincial governments adopt campaign-style enforcement that incorporates the performance in reducing agricultural fires into the local officials' appraisal system.⁴ Officials with the

* Corresponding author. China Academy for Rural Development, School of Public Affairs, Zhejiang University, Hangzhou, China.

E-mail addresses: haihong@zju.edu.cn (H. Hong), kzchen@zju.edu.cn (K. Chen).

¹ Another specific example is that firms may shift production and pollution activities from regulated to unregulated plants, leading to pollution substitution between plants (Chen et al., 2025).

² In China, straw-burning activities usually involve the burning of the stalks of wheat, maize, and rice after harvesting. This is a common practice in many developing countries such as India, Thailand, and the Philippines (Gadde et al., 2009). Throughout this paper, we interchangeably use "agricultural fire", "crop fire", and "straw burning" to refer to the same issue.

³ Burning activities are concentrated in seasons after harvesting. Such intensive burning of crop residues can lead to a substantial escalation in air pollution levels within a few weeks.

⁴ For instance, the upper governments (e.g., central and provincial) require local authorities to specify the corresponding penalties on straw burning activities, and link straw burning prohibition efforts with project approval (essential for local economic growth), total air pollutant emission reduction assessments, so as to motivate the local officials to continuously improve the utilization of straw and most importantly, the prohibition of straw burning.

worst performance may receive serious punishment (Wang et al., 2022). Driven by strong mobilization of the UPSB policy, local officials impose significant monetary and administrative penalties on farmers who burn straws, and resort to costly measures (e.g., relocating government staff in the field for on-site monitoring) to prevent straw-burning activities.⁵

Despite its stringency, the UPSB policy offers limited economic incentives for farmers and potentially distorts the agricultural production process. Since farmers are no longer allowed to burn the residues, they are forced to clear fields through either straw returning or recycling, both of which are time-consuming and labor-intensive. More importantly, the UPSB policy also increases the likelihood of pests and diseases, which can negatively impact agricultural output and productivity. To combat these negative shocks, farmers may adjust their production behavior by using more fertilizers and pesticides. Such changes in factor inputs may trigger environmental externalities in the form of water pollution (Dias et al., 2023; Lai, 2017), resulting in a potential pollution substitution effect (Gibson, 2019; Greenstone, 2003).

To identify the validity and pollution substitution effect of the UPSB policy, we compile a comprehensive dataset from multiple sources. First, we create a grid-level dataset that covers the entire Chinese mainland at a 10 km × 10 km resolution. We then match it with satellite-based observations that measure the number of agricultural fires and air pollution concentration. We also use the satellite-based cropland share to construct a treatment intensity measure of the UPSB policy. Second, we exploit rural household survey data to investigate how farmers respond to the UPSB regulation. The data is drawn from the National Fixed Point Survey (NFPS) dataset, which records detailed agricultural production (e.g., input and output) information. Third, we leverage readings from water monitoring stations to identify the potential effects of the UPSB policy on water pollution.⁶

Our estimation strategy relies on a generalized DiD framework, which exploits two sources of variation. The first is the temporal variation arising from the staggered implementation of the UPSB policy, and the second is cross-sectional variations stemming from differences in *ex-ante* cropland cultivating acreage of each grid cell. Such a design is akin to the general case that combines both continuous treatment with staggered adoption in recent DiD econometric literature (Callaway et al., 2024; de Chaisemartin and D'Haultfœuille, 2024). Conditional on the UPSB policy, our identification assumption is that units with more *ex-ante* cropland should be more exposed to the policy, as burning activities are more prevalent in these areas (Garg et al., 2024). Our main specification compares the relative change in the number of agricultural fires in post-treatment periods relative to the pre-treatment periods, between grid cells with higher cropland shares *ex-ante* versus those with less.

We first estimate the UPSB policy's effect on straw burning activities and air pollution. Our main results show that grids with higher cropland shares in the pre-treatment periods experienced a significantly larger decrease in agricultural fires after the UPSB policy. Specifically, in our preferred specification that includes grid cell fixed effects, county-by-year fixed effects, and treatment-intensity-by-year fixed effects, we identify an approximately 14 percent decrease in agricultural fires after the policy implementation. This reduction corresponds with a decrease in particulate matter (PM_{2.5}), which is in line with findings from existing literature (Guo, 2021; He et al., 2020a). We provide several pieces of evidence to show that political incentives and top-down accountability are the driving forces of the observed reduction in agricultural fires.

⁵ Some anecdotal evidence reveals that the county expenses on local cadres' meals within a single month during the period of high incidence of straw burning (e.g., seasons after harvesting or before cultivating) can even amount to 100,000 RMB (approximately 13,760 USD). See the report from https://www.guancha.cn/politics/2024_03_29_730046.shtml. Accessed at 2025-11-04.

⁶ We will introduce in more detail how we merge the household-level survey data and station-level data into our grid-level dataset in Section 6.

Nevertheless, we also show that high enforcement costs could weaken the effectiveness of the policy.

We then examine how farmers respond to the UPSB policy by adjusting their factor inputs. Using data from multiple sources (e.g., satellite observation, household surveys, and official statistics), we find consistent evidence that farmers increased their fertilizer and pesticide usage after the implementation of UPSB policy. This is in line with our hypothesis that the UPSB policy leads to negative shocks to agricultural production by decreasing soil fertility and increasing the incidence of pest disease.

Lastly, we investigate how the UPSB policy unintendedly leads to increased water pollution, as a consequence of increased fertilizer and pesticide usage. Using both station-level and grid-level data, we find that the UPSB policy leads to worsened water quality, and higher concentrations of chemical oxygen demand (COD) and ammonia nitrogen (NH₃-N), both of which are highly correlated with the intensive usage of chemical fertilizers and pesticides (Lai, 2017). In accordance with the political incentive narratives, we show that the effect of UPSB policy on water pollution is magnified when local officials have higher promotion incentives. Several heterogeneous exercises confirm that our results are indeed driven by the UPSB policy. Specifically, we show that the effects on water pollution are stronger when enforcement costs are lower, fertilizer usage is higher, or precipitation increases (causing surface runoff that carries fertilizers and pesticides into water bodies). In addition, we conduct a series of robustness checks to ensure that our results are not from spurious correlations. In particular, we show that our results are robust to (1) the exclusion of other confounding factors such as change in regulation intensity and industrial water pollution, (2) using an upstream-downstream specification, and (3) adopting alternative measures of water pollution, such as the occurrence of algal bloom.

While conducting a rigorous and comprehensive cost-benefit analysis of the UPSB policy is beyond the scope of this paper, we provide a rough estimate of the net benefits based on our empirical findings and several conservative assumptions. Specifically, we estimate that the benefit of the UPSB policy through improved air quality is 34.11 billion CNY or 5.25 billion USD, while the enforcement cost of the UPSB policy is estimated to be 10.26 billion CNY or 1.60 billion USD. A simple comparison of the benefits and costs leads to the conclusion that the UPSB policy is cost-effective. However, if we further consider the production cost (e.g., input adjustment and potential output losses) and the related environmental consequences, the adjusted health benefit would be 20.02 billion CNY or 3.08 billion USD, and the adjusted policy cost would be 35.98 billion CNY or 5.54 billion USD. Adjusting for the policy benefits and costs leads to an overall negative gain of the UPSB policy, which amounts to a total cost of 15.96 billion CNY or 2.46 billion USD.

This paper makes three contributions to the literature. First, we speak to the burgeoning literature that discusses how to effectively reduce straw burning activities (Cao and Ma, 2023; Dipoppa and Gulzar, 2024; He et al., 2020a; Jack et al., 2025; Nian, 2023). Specifically, this strand of literature discusses both the role of providing economic incentives, as well as leveraging direct regulations, in reducing agricultural fires. For instance, Cao and Ma (2023) and Nian (2023) find that the entry of biomass power plants can reduce straw-burning activities in plant-nearby regions. They also discuss the role of command-and-control policies and find limited evidence that environmental regulations exert positive effects in curbing agricultural fires. Using exogenous change in wind direction, Dipoppa and Gulzar (2024) establish that leveraging bureaucrat incentives can effectively reduce agricultural fires in India. We contribute to this strand of literature by providing comprehensive evidence on the effectiveness and unintended consequences of China's nationwide straw burning bans. This insight aligns with findings from Dipoppa and Gulzar (2024) on the role of political motivation but provides new evidence for the role of centralized policies and cross-regional enforcement in China.

Second, we contribute to studies that investigate the pollution substitution effect in environmental regulations. The empirical evidence on

Province for its inadequate enforcement of straw-burning regulations.⁹ The Henan provincial government soon implemented the UPSB policy in the following year. Other factors, such as the *ex-ante* pollution level, may likely drive the implementation of the UPSB policy as well. We will return to the issue of potential endogenous policy adoption when we formally present our research design.

Unlike previous practices that only relied on regular enforcement to curb agricultural fires, the UPSB policy explicitly mandates that no burning activities are allowed within the entire administrative territory. Specifically, provincial governments are responsible for campaign-style enforcement, where provincial leaders initiate enforcement campaigns and allocate resources (e.g., personnel, funding, etc.) to the county officials, who are fully mobilized to enforce the UPSB policy and are held accountable if they fail to fulfill the pre-specified mandated targets (Wang et al., 2022).

For instance, the Henan Province declares that each additional fire point detected by satellite results in a fine of 50,000 RMB (approximately 7800 USD) for the county government. County officials face admonishment and censure from provincial leaders if more than five fire points are detected. For counties that fail to enforce the policy effectively, the provincial government will impose restrictions on the approval of construction projects, which are crucial for economic development.¹⁰ In addition to the strict accountability of local officials, the Henan government has invested in resources and advanced technologies to ensure successful policy implementation. The province installed a total of 19,262 cameras for real-time monitoring and established 35,343 emergency response teams to achieve full coverage of agricultural land.¹¹ These practices are shown to be effective. During the burning season of 2017 (late September to late November), no fire points were detected by the satellite.

Shandong province is another agricultural province that adopts strict accountability mechanisms to regulate straw-burning activities. Local officials in agricultural counties are subject to suspension or censure for failing to prohibit open burning. For example, in the first half of 2020, only 14 agricultural fire points were detected in total, yet more than 50 local leaders were punished.¹² In China's political system, such accountability is typically enforced through intra-party supervision, where officials may face punishment such as warnings, removal from party positions, or other administrative penalties, which are detrimental to their future political careers. This incentivizes local officials to enforce the UPSB policy with full dedication, as minor oversight may result in severe consequences.

However, the effectiveness of campaign-style enforcement comes with high enforcement costs. Monitoring the numerous small-scale farmers who have the potential to burn crop residues is costly (Cao and Ma, 2023), requiring local officials to mobilize a mass of grassroots cadres, which creates additional fiscal pressure on local governments. Anecdotal evidence reveals that in some counties implementing the UPSB policy, monthly fiscal expenditures on cadres' catering alone exceed a hundred thousand RMB, and the effects of curbing burning activities are merely moderate.

After the rollout of the UPSB policy, the NBZ policy remains in effect. Although the central government banned the NBZ policy in 2015, some provinces still emphasize intensifying regulation efforts in these key areas, such as those surrounding airports, expressways, and railways. However, local officials may only be willing to resort to more efforts in

areas that are more easily noticed by their superiors. In order to supervise local officials, provincial leaders (sometimes the prefectural leaders) often do random inspections to examine whether local agents are devoted to enforcing the given task. Expressways are the major transportation network that facilitates access to random spots, given their convenience and concealability, and lack of time constraints (relative to railways). As a result, local officials are more incentivized to enforce strict regulations in areas near expressways, while devoting less effort to other NBZs that may not be easily supervised.

2.3. Potential consequences of the UPSB policy

Despite its efficiency in curbing agricultural fires, the UPSB policy also imposes additional costs on farmers. As mentioned above, farmers need to devote additional resources to manage crop residues, with straw returning being the most prevalent way.¹³ Nonetheless, straw returning has significant drawbacks.

First, it is susceptible to pests and diseases. In the returning process, germs and pests hidden in the straws are returned as well, affecting crop yields and quality. For example, the Department of Agriculture of Zhejiang Province states that although the province's UPSB policy has improved air quality, it has brought adverse effects, such as exacerbating diseases like rice blast. In Changsha, Hunan Province, agricultural statistics show rice planthoppers (a typical rice pest) in 2022 were 14.8 times higher than those before the UPSB policy in 2017.

Moreover, it takes time for straws to decompose and be absorbed by the soil. This matters especially in northern regions with lower temperatures, where the process of decomposing crop residues into fertilizer is slower. If straws are not fully decomposed, they can cause crop roots to rot. A critical example is Heilongjiang Province, the northernmost region of China. Even though the province has ramped up enforcement against straw burning, farmers still burn straw secretly before the sowing season, as the costs of straw returning are way larger than burning.¹⁴ In addition, once the amount of straw returned to the field exceeds the soil's optimal capacity, decomposition time will be significantly prolonged, and the over-accumulation of straw can result in excessive soil erosion and land degradation.

Given the potential negative shocks brought by the UPSB policy and the reluctant adoption of straw returning, farmers could adjust their production behavior to mitigate the losses in production. The most direct way is to adjust the farmland (i.e., the production scale). Faced with environmental regulations, farmers may reduce food production and seek urban employment. However, farmers who are less able to make such adjustments may change their production inputs to offset negative shocks.¹⁵ Typically, to combat pests and diseases, farmers may use more fertilizers and pesticides. While the UPSB policy curbs air pollution from burning crop residues, the overuse of fertilizers and pesticides can lead to additional negative externalities, such as water pollution (Lai, 2017). Eventually, this pollution substitution effect caused by farmers' adjustment behaviors may undermine the UPSB

⁹ See document from https://www.mee.gov.cn/gkml/hbb/bgth/201409/t20140918_289253.htm. Accessed at 2025-11-04.

¹⁰ See report from https://www.gov.cn/xinwen/2015-05/30/content_2870801.htm. Accessed at 2024-04-07.

¹¹ See report from https://www.gov.cn/xinwen/2017-11/22/content_5241523.htm. Accessed at 2024-04-07.

¹² See report from <https://news.iqilu.com/shandong/yuanchuang/2020/0712/4588871.shtml>. Accessed at 2025-11-04.

¹³ According to the 2021 data, among crop residues that are not burned away, 62.5% are returned to the farmland, while the remaining 37.5% are recycled for other usage. See the report from https://m.thepaper.cn/baijiahao_20861205. Accessed at 2025-11-04.

¹⁴ In 2023, county governments were fined for 178 million RMB (approximately 28 million USD) due to ineffective enforcement in regulating straw burnings. Within the same year, 185 farmers were detained for engaging in open burning activity, and over 400 local officials were held accountable. See more details from <https://www.163.com/dy/article/HVBGFSSM0556165R.html>. Accessed at 2025-11-04.

¹⁵ For instance, individuals who are more specialized in food production or are unable to make occupation adjustments due to labor market friction, which is more prevalent in developing countries due to the incomplete labor market and dual economic structure. In China, institutional barriers, such as the Hukou system, also impede the process of rural-urban migration.

policy's intended benefits.

Despite the importance of the potential pollution substitution effect in the implementation of UPSB policy, empirical investigation into these effects remains limited. We proceed to further examine the effects and consequences of the UPSB policy and attempt to provide a comprehensive evaluation of its costs and benefits.

3. Data

3.1. Research sample

We assemble information on agricultural fires, air pollution, water pollution, and agricultural inputs from various sources and construct grid-level panel data with cells of $10\text{ km} \times 10\text{ km}$ resolution covering all of China's mainland from 2001 to 2019.¹⁶ The use of such disaggregated data possesses several merits. First, our primary measure of the treatment (i.e., adopting the UPSB policy) is at the provincial level, which may be confounded by unobserved time-varying factors. The grid-level data allows us to include grid fixed effects to control for unobserved heterogeneities. Second, by defining our treatment intensity (i.e., cropland shares) at the grid level, we can capture more variations in the data. This approach allows us to introduce cross-sectional variation and control for county-year fixed effects to address the selection bias of the implementation and enforcement stringency of the UPSB policy. This is particularly crucial, as county leaders serve as the local agents responsible for regulating straw burnings.

3.2. Data source

Agricultural fires data. The fire point data used in our paper is sourced from NASA's MODIS aboard the Terra and Aqua satellites, which have been frequently used in recent studies on agricultural fires (Cao and Ma, 2023; Nian, 2023). These satellites pass over China twice daily, typically occurring between 10 a.m. and 3 p.m., and between 9 p.m. and 2 a.m. China Standard Time. MODIS sensors identify fires using a contextual algorithm that detects the strong emission of mid-infrared radiation from fires and reports their longitude and latitude. The satellites started to record fire points in November 2000, and we therefore use data from 2001 to 2019. We identify agricultural fire by leveraging land cover data from the China Land Cover Dataset (CLCD), a remotely sensed product providing nationwide land type classifications at 30-m resolution from 1990 to 2020 (Yang and Huang, 2021). We match the fire point data to the land cover raster and define fires as agricultural fires if they occur within cropland. Non-agricultural fires (i.e., fires that occur outside cropland) are also calculated, with both types aggregated at the grid-year level.

Cropland acreage data. Our treatment intensity is defined as the total cropland share of major straw-producing crops within each grid cell. These crops, including wheat, maize, and rice, collectively contribute to over 80% of total straw output. We utilize data from Luo et al. (2020), accessible through the National Ecosystem Science Data Center (NESDC), to calculate cropland share.¹⁷ The dataset covers the spatial distribution of maize, wheat, and rice cultivation in China from 2000 to 2019, with a resolution of 1 km. To avoid reverse causality in our identification strategy, we only use data from 2001 to 2010, prior to the implementation of the UPSB policy.¹⁸ We aggregate the cropland acreage dedicated to these three crops to establish the total cropland

acreage, dividing it by 100 (the area of the 10 km grid cell) to derive the cropland share. While most of the cropland shares range between zero and one, they may exceed one in some instances due to crop rotation and replanting.¹⁹

Satellite air pollution data. We obtain ground-level PM_{2.5} data for each grid cell from the NASA Socioeconomic Data and Applications Center (SEDAC).²⁰ The dataset combines AOD retrievals from multiple satellite algorithms and exploits the GEOS-Chem chemical transport model to relate the total column measure of aerosol to near-surface PM_{2.5} concentration. Calibration is performed using Geographically Weighted Regression (GWR) to produce the final products. We aggregate the raw raster data of PM_{2.5} concentrations, originally captured at a 1 km resolution, to our 10 km grid-level data and compute the annual average of PM_{2.5} concentration for each grid.

Chemical fertilizer usage data. We employ annual grid-level nitrogen fertilizer use data from Yu et al. (2022).²¹ Based on several official fertilizer data sources of China and FAO, and spatial information of cropland acreage, crop-specific planted acreage, and crop rotation, the dataset provides historical nitrogen fertilizer use at a $5\text{ km} \times 5\text{ km}$ resolution, covering periods from 1952 to 2018. We then aggregate the fertilizer data into our grid-level panel, ranging from 2001 to 2018. With the data in hand, we can evaluate how the UPSB policy leads to unintended use of fertilizer.

Water pollution data. We use two datasets to examine the UPSB policy effects on water pollution induced by higher fertilizer use. The first is automatic weekly surface water pollution data released by the Chinese National Environmental Monitoring Center. The water pollution data spans from 2004 to 2018, and covers 148 water quality monitoring stations for major rivers, lakes, and reservoirs. Appendix Figure A1 shows the spatial distribution of water quality monitoring stations. We select water quality grade, ammonia nitrogen (NH₃-N), and chemical oxygen demand (COD) concentration as the main indicators for measuring water pollutants. Water quality grade, a categorical variable that ranges from 1 to 6, measures the overall water quality based on different water pollutant indicators including the pH scale, dissolved oxygen, COD, NH₃-N, and total phosphorus. The higher the water quality grade, the worse the water quality. NH₃-N is a measure of the total amount of ammonia (NH₃) and ammonium ions (NH₄⁺) present in the form of nitrogen in the water body. Chemical fertilizer application is one of the main sources of NH₃-N. The COD measures the amount of oxygen consumed to chemically oxidize organic water contaminants to inorganic end products in water. We first aggregate the weekly water pollution data into annual averages and then exploit two strategies to match station-level data to our grid-level data. The first strategy aggregates grid-level data to the station level, while the second strategy does the opposite. Both strategies have their pros and cons, which we will illustrate in further detail when investigating the policy effects on water pollution.

Fertilizer use is also associated with the occurrence of algal blooms due to excessive nutrient leaching (Taylor and Heal, 2021). Our second dataset that measures water pollution uses algal bloom data of lakes from Wang et al. (2023),²² which is generated using MODIS satellite observations. This dataset provides three key indicators in terms of bloom occurrence, potential occurrence period, and maximum bloom extent for 103 China freshwater lakes from 2003 to 2020. The bloom

¹⁶ Since our data is derived from satellite observations with different resolutions. We choose the $10\text{ km} \times 10\text{ km}$ resolution to ensure that all satellite data can be mapped to our grid cell coherently.

¹⁷ See more details of the data from <http://www.nesdc.org.cn/sdo/detail?id=627dfc4b7e28172589c2df9b>.

¹⁸ Our results are robust to an alternative definition of the treatment intensity, as examined in the robustness checks.

¹⁹ The results are not affected if we either discard observations with cropland shares greater than one or instead replace them with one.

²⁰ The data is from (Hammer et al., 2020, 2022). See more details from <https://sedac.ciesin.columbia.edu/data/set/sdei-global-annual-gwr-pm2-5-modis-misr-seawifs-aod-v4-gl-03>

²¹ The data is from Yu et al. (2022). See more details from <https://essd.copernicus.org/articles/14/5179/2022/>.

²² The data is from Wang et al. (2023). See more details from <https://doi.org/10.1029/2022WR033340>.

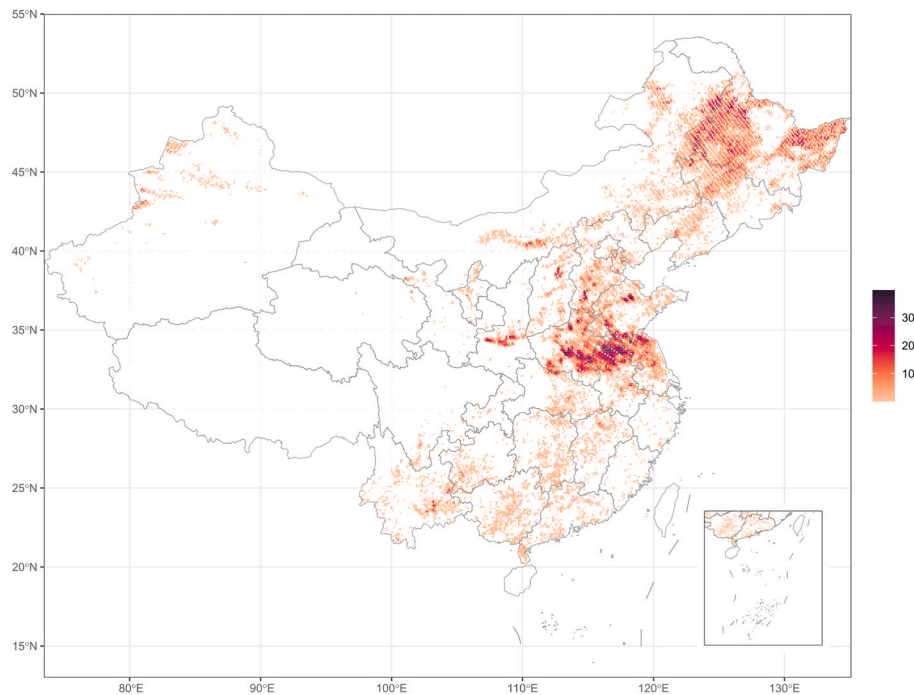


Fig. 1. Satellite-detected agricultural fires for each grid during 2001–2019

Notes: Colored red points represent the number of agricultural fires for each grid, with darker shades indicating higher levels of burning activity. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

has adopted the UPSB policy in year t , outlined in Equation (1):

$$y_{ipt} = \beta Post_{pt} + X_i \times T_t + Z_{it}\theta + \delta_i + \eta_t + \epsilon_{ipt} \quad (1)$$

Where y_{ipt} represents the number of agricultural fires detected in grid i at year t of province p , or its inverse hyperbolic sine (IHS) transformation, capturing the level or the proportional change in the treatment effects.²⁶ The policy variable, $Post_{pt}$ is defined at the province-year level, which equals to 1 for years after the implementation of UPSB policy and 0 otherwise. X_i and Z_{it} are two vectors denoting geographic controls and weather controls, respectively. The time-invariant geographic controls (e.g., slope, distance to the airport, etc.) are interacted with linear year trends to allow for their temporal effects. We also include grid fixed effects and year fixed effects to absorb grid-specific characteristics and time-varying common shocks. This setting compares the average changes in the number of agricultural fires in the treated provinces relative to the control provinces.

Since treatment status is only measured at the province-year level in Equation (1), this parsimonious specification cannot account for within-province variations that may confound our identification, potentially leading to omitted variable bias or simultaneity issues. For instance, the construction of rural roads could increase the likelihood of crop residue burning by inducing labor exits (Garg et al., 2024). Such omitted variable bias may lead to an underestimation of the true policy effects. Simultaneity issues may arise from other concurrent policies that may affect the burning activities. One example is the land titling reform (Bu and Liao, 2022; Liu et al., 2023), which has been widely proven to

facilitate land consolidation by ensuring land property rights.²⁷ Land consolidation facilitates the use of machinery to recycle crop residues and therefore may reduce straw burning activity, potentially leading to overestimation.

To address the above concerns, we refine our model by incorporating grid-level variations in the cropland share. Specifically, we augment the parsimonious specification in Equation (1) by interacting the policy variable $Post_{pt}$ with the grid-specific cropland share, specified as follows:

$$y_{icgpt} = \beta Post_{pt} \times Cropland_i + X_i \times T_t + Z_{it}\theta + \delta_i + \eta_{ct} + \zeta_{gt} + \epsilon_{icgpt} \quad (2)$$

Where $Cropland_i$ denotes the average cropland share of wheat, maize, and rice in grid i . This interaction allows us to exploit within-province variations and use more granular fixed effects to account for unobserved confounders. In particular, we additionally control for the county-year fixed effects η_{ct} to absorb county-specific time-varying heterogeneities. Since the enforcement of UPSB policy is largely relegated to the county officials, we control for the county-year fixed effects to restrict the comparison within the same county in the same year.²⁸ Standard errors are two-way clustered at the grid level and the county-year level to account for correlations across years within the same grid and correlations across all grids in the same county and same year, following Cameron et al. (2011).

We adopt such continuous measures for treatment intensity for two reasons. First, the original measure of treatment is defined at the provincial level, which may be rather coarse and underpowered. The continuous measure allows us to capture more variation in the data and thus leads to more precise estimates. Second, the provincial level measurement may be confounded by other unobserved heterogeneity, potentially biasing our estimates. By exploiting the grid-specific

²⁶ Since the vast majority of grids has a zero agricultural fire, we use IHS to avoid missing these observations. We do not use more conventional approach like log+1 transformation as it may be problematic to interpret the estimated coefficients as percentage changes and there could be potential bias in such transformation (Chen and Roth, 2024). However, as pointed out by Chen and Roth (2024), using IHS transformation may not necessarily alleviate the concerns, and we address such issue in our robustness checks by using the Poisson pseudolikelihood regression as alternative specification.

²⁷ The timing of land titling reform is decided by the provincial governments, but county governments hold discretionary power to decide precisely to carry out the reform (Bu and Liao, 2022).

²⁸ The policy variable itself is absorbed by the county-year fixed effects and is thus not included in the regression.

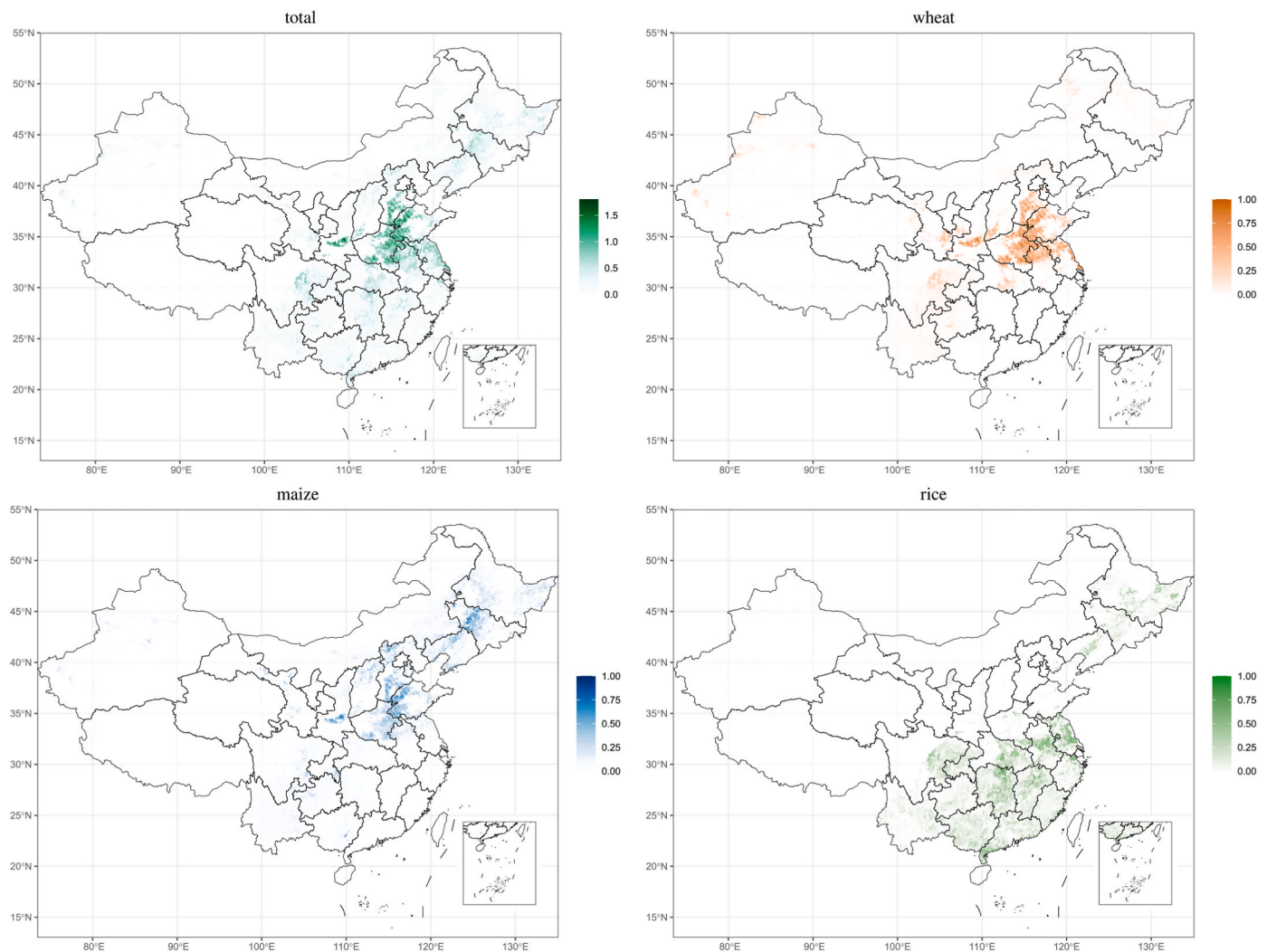


Fig. 2. Average cropland shares for wheat, maize, and rice

Notes: This figure plots the average cropland share for each grid cell, from 2001 to 2010. The top left panel presents the sum of the three crops, while the rest three panels display the cropland share for each specific crop.

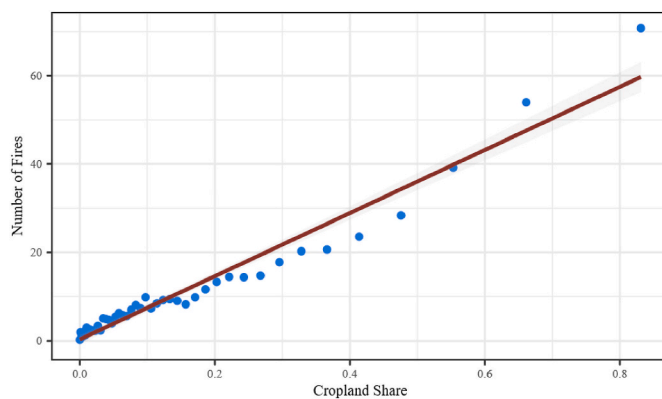


Fig. 3. Binscatter plot on the relation between the number of fires and cropland share

Notes: This figure displays the binscatter plot between the number of agricultural fires and the cropland share. Additionally, it includes the fitted unconditional regression line and the corresponding 95% confidence intervals, shaded in grey.

variation in cropland acreage, we are able to partial out a wide range of

within-province confounders with the inclusion of more granular fixed effects, leading to more credible inference.

The above generalized specification in Equation (2) can be interpreted as a variant of the canonical DiD design, with both treatment timing and treatment intensity varying across groups (Callaway et al., 2024; de Chaisemartin and D'Haultfœuille, 2024). Typically, the decomposition in Callaway et al. (2024) indicates that the TWFE estimator is, in essence, using the above-average treatment intensity units as the effective treatment group and below-average treatment intensity units as the effective control group.²⁹ This raises concerns that the TWFE estimator may be biased if the effective treatment groups and control groups have divergent trends even in the absence of the UPSB policy, which is plausible as the economic activities could vary in grids with different cropland shares.

To alleviate such concerns, we further include the (effective) group by year fixed effects ζ_{gr} . Specifically, we classify grids into high versus low treatment intensity groups based on their predetermined cropland

²⁹ The comparison between the two groups (i.e., grids with above-average cropland share and below-average cropland share), across pre- and post-treatment periods, yields the estimated results in our generalized DiD design. However, this approach requires additional identification assumptions for clear causal interpretation.

share, and then interact the group variable with a full set of time dummies to control for potential diverging trends between the effective treatment and control groups. Conditioning on these fixed effects, we can leverage the narrowed within county-year and group-year variations for identification, leading to more credible and robust results. The coefficient β measures the relative change in the number of agricultural fires in post-treatment periods relative to the pre-treatment periods, between grid cells with higher cropland shares *ex-ante* versus those with less. We adopt equation (2) as our workhorse model and use equation (1) for additional support and comparisons.

Additionally, we also explore whether our estimates are sensitive to the potential negative weighting issues highlighted in recent econometric literature (Callaway et al., 2024; Callaway and Sant'Anna, 2021; de Chaisemartin and D'Haultfœuille, 2024; 2020; Goodman-Bacon, 2021; Sun and Abraham, 2021). Negative weighting issue arises in staggered adoption designs with heterogeneous treatment effects (de Chaisemartin and D'Haultfœuille, 2020; Goodman-Bacon, 2021), and recent work extends this issue to continuous treatment designs (Callaway et al., 2024; de Chaisemartin and D'Haultfœuille, 2024).³⁰ Given our model's incorporation of staggered adoption and continuous treatment, the presence of the negative weighting problem could substantially bias our estimates. To examine whether the presence of negative weights significantly biased our results, we adopt the estimator proposed by de Chaisemartin and D'Haultfœuille (2024) that is robust to both continuous and staggered treatment (de Chaisemartin and D'Haultfœuille, 2023).³¹

Finally, to provide visual support for our identification assumption (i.e., parallel trends), we use the following event study specification to scrutinize the presence of pre-trends:

$$y_{icgpt} = \sum_{k=-6, k \neq -1}^4 \beta_k Post_{pt}^k \times Cropland_i + X_i \times T_t + Z_{it}\theta + \delta_i + \eta_{ct} + \zeta_{gt} + \epsilon_{icgpt} \quad (3)$$

Here, we replace the policy shock indicator $Post_{pt}$ with a set of time dummies indicating the periods relative to the UPSB policy implementation. The year prior to the policy enforcement (i.e., $k = -1$) is omitted as the reference year. The parameters β_k s thus measure changes in agricultural fires in high versus low treatment intensity groups, between the k th period relative to the UPSB policy implementation, and the period prior to its implementation.

5. Impacts on agricultural fires and air pollution

5.1. Main results

Baseline estimates. Table 1 presents our baseline results using the generalized DiD estimation specified in equation (2). Columns (1) to (3) examine the effects of UPSB policy on the number of fires, while columns (4) to (6) examine the effects on the IHS transformation of fires. Columns (1) and (4) include grid cell fixed effects, county-by-year fixed effects, and the (effective) group-by-year fixed effects. In columns (2) to (3) and (5) to (6), we sequentially add geographic and weather controls to

³⁰ Specifically, in our context, the negative weighting issues stemming from the staggered adoption should be of less concern as we only have a fairly small group with different treatment timings, and there is a vast majority of never never-treated group. And the inclusion of county by year fixed effects further alleviates concerns of “forbidden comparison” that compares the outcome between early treated groups with later treated groups.

³¹ It is worth noting that, although our fixed effects model may be troubled by the potential weighting issues, we prefer it as our workhorse model due to its flexibility, that allows us to examine the heterogeneous effects of the UPSB policy, which could provide further insights from a policy-relevant perspective. Reassuringly, our empirical results established below suggest that the negative weight does not lead to substantial bias in our findings.

examine the stability of our coefficients to observed grid characteristics. The results show minimal changes, with all estimated coefficients significant at the five percent level.³²

In our preferred specifications, which include the full set of fixed effects and control variables (columns (3) and (6)), we find that the UPSB policy can effectively curb agricultural fires in areas with high cropland shares. Specifically, given that the average grid in our sample has an 8 percent cropland share prior to the UPSB policy, the estimate in column (3) suggests that the introduction of the UPSB policy is associated with a decrease in the number of agricultural fires of about 0.83 ($= -10.359 \times 0.08$), which is approximately 14.4% drop relative to the sample mean (5.805). Alternatively, moving across the interquartile range of the cropland share (a change of about 0.07) implies a decrease of 0.77 in the number of agricultural fires after the implementation of the UPSB policy. The results remain robust when using the IHS transformation of agricultural fires as the dependent variable. Specifically, a one percentage increase in the cropland share would lead to a 14.6% ($= \exp(-0.158) - 1$) decrease in the agricultural fire.³³ Our estimated effects are comparable to Cao and Ma (2023), who documented that the entry of biomass power plants (BPP) can decrease the number of agricultural fires by 14%. However, since their estimates are more localized (only within a certain radius surrounding the plant), the overall estimated magnitude of the UPSB policy should be larger than that obtained by Cao and Ma (2023).³⁴

Another concern with our baseline empirical design is that we are comparing the relative change in straw-burning activities between grids with a higher cropland share and grids with a lower cropland share. Though less plausible, the negative coefficients may reflect a relative increase in crop fires in grids with a lower cropland share. To ensure that this is not the case, in Appendix Table A6, we run equation (1) separately for grids with high cropland share (above mean) and low cropland share (below mean) to pin down the absolute effects of the UPSB policy. As expected, we find that the decrease in crop fires is concentrated in grids with a higher cropland share, and we do not find any significant policy effect on crop fires in grids with a lower cropland share. This confirms our interpretation that the UPSB policy mainly exerts its effects by reducing crop fires in grids with a higher cropland share (and thus more incidences of burning activities).

Although we've controlled for the initial cropland share in each grid, our specification does not allow us to identify whether the decrease is driven by the extensive margin (i.e., decrease in cropland share) or the intensive margin (i.e., decrease in burning intensity). If the decrease in crop fires is completely driven by the extensive margin, then conditional on the *ex-post* cropland share should fully absorb our estimated coefficients. If the decrease is driven by the intensive margin, then conditional on the *ex-post* cropland share should have no effect on our estimates. We perform this test in Appendix Table A7, where we find that the estimated coefficients are not affected by the inclusion of the *ex-post* cropland share, suggesting that the decrease in crop fires is mainly driven by the intensive margin.

A direct effect of the reduction in agricultural fires is improved air quality. We thus explore how the UPSB policy contributes to decreased air pollution. The results are reported in Table 2, where we estimate

³² We also estimate the policy effect using the parsimonious specification outlined in equation (1), as reported in Appendix Table A5. We find similar results that the UPSB policy can effectively reduce agricultural fires. However, the estimated results should be interpreted with caution due to potential within-province confounding factors.

³³ Rigorously, the elasticity calculated should be $\beta \times \frac{\sqrt{y^2+1}}{y}$ when the dependent variable is IHS-transformed. In this case, our estimates imply a reduction of fires by a 14.8% (Bellemare and Wichman, 2019).

³⁴ We also empirically examine the role of the BPP entry and compare its effects with our baseline findings in Appendix Table B8. See Appendix B for further discussion.

Table 1

The effects of UPSB policy on agricultural fires.

Dep. Var.	(1)	(2)	(3)	(4)	(5)	(6)
	Number of Fires			IHS (# of Fire)		
<i>Post</i> × <i>Cropland</i>	−10.328** (4.160)	−10.343** (4.162)	−10.359** (4.165)	−0.155** (0.069)	−0.158** (0.069)	−0.158** (0.069)
Observations	1,717,358	1,717,358	1,716,717	1,717,358	1,717,358	1,716,717
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
County by Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Group by Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo Controls	No	Yes	Yes	No	Yes	Yes
Weather Controls	No	No	Yes	No	No	Yes
Dep. Var. Mean	5.805	5.805	5.805	0.125	0.125	0.125
Adjusted R-squared	0.539	0.539	0.540	0.546	0.546	0.546

Notes: The unit of observation is 10 km × 10 km grid cells. This table shows that the universal prohibition on straw burning (UPSB) policy can reduce agricultural fires in areas with a high cropland share, using a generalized difference-in-differences strategy. The dependent variables are the number of fires and the IHS transformation of agricultural fires. *Post* is an indicator for years after the policy implementation. *Cropland* is the cropland share for each grid cell. Geographic controls include slope, ruggedness, elevation, distance to the county border, county center, and provincial center; distance to rivers, and the number of rivers; distance to airports, distance to expressways, distance to railways, and distance to national roads and provincial roads. Weather controls include temperature, precipitation, humidity, sea level pressure, wind speed, and wind direction. Standard errors in parentheses are two-way clustered at the grid level and the county-year level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

Table 2The effects of UPSB policy on PM_{2.5}.

Dep. Var.	(1)	(2)	(3)	(4)
	log(PM _{2.5})			
<i>Post</i>	−0.221*** (0.080)	−0.214*** (0.055)		
<i>Post</i> × <i>Cropland</i>			−0.012** (0.006)	−0.012** (0.006)
Observations	1,755,780	1,755,200	1,753,682	1,753,102
Grid FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	–	–
County by Year FE	–	–	Yes	Yes
Group by Year FE	–	–	Yes	Yes
Geo Controls	No	Yes	No	Yes
Weather Controls	No	Yes	No	Yes
Dep. Var. Mean	3.045	3.045	3.045	3.045
Adjusted R-squared	0.892	0.914	0.989	0.989

Notes: The unit of observation is 10 km × 10 km grid cells. This table reports the effects of the UPSB policy on air pollution, using both parsimonious and baseline specifications. The dependent variable is the log transformation of PM_{2.5}. *Post* is an indicator for years after the policy implementation. *Cropland* is the cropland share for each grid cell. Geographic controls include slope, ruggedness, elevation, distance to the county border, county center, and provincial center; distance to rivers, and number of rivers; distance to airports, distance to expressways, distance to railways, distance to national roads and provincial roads. Weather controls include temperature, precipitation, humidity, sea level pressure, wind speed, and wind direction. Standard errors in parentheses are two-way clustered at the grid level and the province-year level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

both the parsimonious specification and the baseline specification. The parsimonious specification suggests that the UPSB policy is associated with a significant reduction in PM_{2.5} by over 20%. However, this should be interpreted with caution, as other environmental policies aiming to reduce ambient air pollution, such as the Action on Pollution Prevention and Control Policy, and the gradual rollout of air pollution monitoring stations across cities, were also in effect at the time. Our baseline specification offers a more credible interpretation, suggesting that the

UPSB policy leads to an approximately 0.1 percent reduction in PM_{2.5} for a grid with an average cropland share.³⁵

Event study. The identification of our generalized DiD design hinges on the parallel trends assumption, which requires that, in the absence of the UPSB policy intervention, there should be no differential trends between the high- and low-treatment intensity groups. It is important to note that the parallel trends assumption does not necessitate that the treatment intensity (i.e., cropland share of each grid cell) be exogenous. Instead, it only requires that trends across groups with different treatment intensities evolve similarly. To provide support for the assumption, we exploit the event study approach specified in equation (3) and visualize the estimated coefficients with corresponding 95% confidence intervals in Fig. 4. The left panel plots the dynamic effects on the number of fires, while the right panel plots the effects on the IHS-transformed number of fires. Both panels exhibit similar patterns. In the pre-treatment periods, there are some upward trends; however, the estimated coefficients are jointly insignificant and fluctuate around zero. After the implementation of the UPSB policy, the number of agricultural fires decreased with a one-year lag. This is plausible as the burning of crop residues is often a seasonal activity. Within the same year, the burning could occur before the policy documents are released, and thus, the policy can only reduce crop fires in the following years. Indeed, in the first year after thereafter of policy implementation, we estimate persistent and significant negative effects, suggesting that the UPSB policy is effective in curbing agricultural fires in both the short term and long term. We also present the event study estimates for the parsimonious specification in Appendix Figure A3. Consistent with our main findings, there is no evidence of pre-trends.

We further address the negative weighting issues in our generalized setting, where treatment effects may be heterogeneous in different units (Callaway et al., 2024). We adopt the DiD₁ estimator proposed by de Chaisemartin and D'Haultfoeuille (2024), which is robust to both staggered adoption and continuous treatment. Specifically, the DiD₁ estimator is a weighted average, across different time periods and possible values of treatment intensity, of 2 × 2 DiD building blocks that compare the outcome evolutions in grids with the same treatment intensity, between treatment and control groups (either pure control or not-yet treated groups), before and after the treatment. By carefully selecting

³⁵ This estimate, however, should be interpreted with caution as it does consider the strategic behavior of local officials. For instance, by regulating straw burning activities, local officials may benefit from relaxing the regulation over pollution firms, which is more essential for the local economic growth, compared with agricultural production.

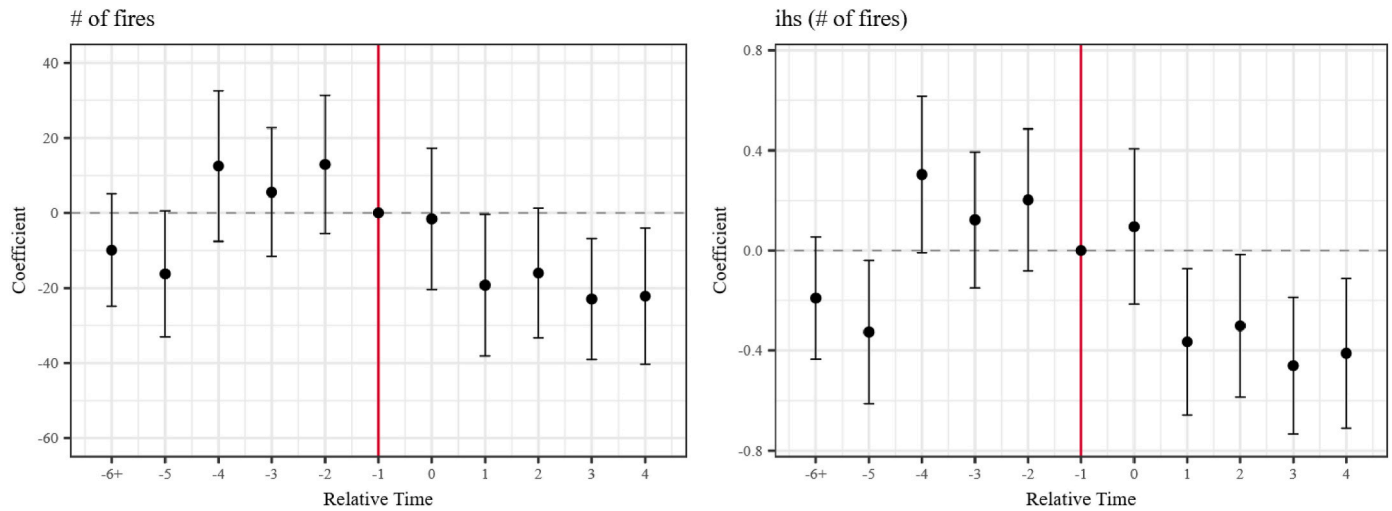


Fig. 4. Event study estimates of the UPSB policy on agricultural fires

Notes: This figure plots the estimated event-study coefficients of the impact of the UPSB policy on agricultural fires. The left panel plots the effects on the absolute number, while the right panel plots the effects on the IHS-transformed number of fires. The unit of observation is 10 km $\times$ 10 km grid cells. Regression specification is presented in Equation (3). Coefficient estimates are plotted with the 95% confidence interval. All regressions include geographic and weather controls. Standard errors are two-way clustered at the grid level and the county-year level.

the comparison groups, the DiDi estimator avoids the pitfalls of negative weighting issues in conventional designs. Fig. 5 presents the corresponding results. Reassuringly, our event study estimates remain robust after accounting for treatment effects heterogeneity. We do not detect any significant pre-trends that may violate the parallel trends assumption, and the dynamic policy effects remain negatively significant.

Although the DiDi estimator does not allow for an aggregation of treatment effects across different treatment periods as in Callaway and Sant'Anna (2021), we perform a simple aggregation to inform the implied estimated effects after correcting the negative weighting problem. Specifically, we calculate the weighted average of the post-treatment coefficients in Fig. 5, with the weights being the number of observations used in the estimation of each treatment effect. This yields an estimated treatment effect of -12.815 (left panel) and -0.1489 (right panel), which is similar to our fixed effects estimates in Table 1.³⁶ This alleviates concerns that negative weights may substantially bias our results and presents supportive evidence that our fixed-effects estimator, at least to some extent, is valid in recovering the true policy effects.

In Appendix Figure A4, we also present the corresponding event study estimates for PM_{2.5}. Consistent with results from Figs. 4 and 5, we find no evidence of significant pre-trends while identifying a persistent decline in air pollution, suggesting that the regulation against straw burning is effective in reducing air pollution.

Robustness. To streamline our empirics, we refer interested readers to Appendix B for additional evidence that supports the validity and robustness of our baseline findings. Specifically, Appendix B scrutinizes the sensitivity of our baseline results to different specifications and discusses several potential confounding factors that may affect the estimated effects (e.g., straw recycling as in He et al. (2020a), the entry of biomass power plant as in Cao and Ma (2023) and Nian (2023), and other concurrent environmental policies). We also try out some alternative identification strategy, e.g., Difference-in-Discontinuities (DiDisc) that evaluates the UPSB policy at the provincial border. The results remain robust.

Before proceeding to investigate how farmers may respond to the

policy shock and the corresponding consequences, we briefly discuss the heterogeneous effects of the UPSB policy on agricultural fires to shed some light on the potential mechanisms through which the UPSB policy comes into effect.

5.2. Heterogeneity

Having established a robust causal relationship between the UPSB policy and the reduction in agricultural fires, this section explores the potential mechanisms driving our results. Combined with the campaign-style nature of the UPSB policy that mobilizes local officials to enforce regulations, we hypothesize that political incentives and top-down accountability are the main driving forces behind the observed reduction effect. Additionally, enforcement costs and local officials' strategic responses associated with implementing the UPSB policy may mediate our baseline estimates. We explore the role of these factors further below.

Political incentives. To effectively implement the UPSB policy, the provincial governments fully mobilize the local officials (i.e., county and prefecture leaders). As documented in the Background section, local leaders face penalties and accountability if they fail to achieve the policy targets. These penalties can significantly impact the careers of local leaders, especially those with strong incentives for promotion. We, therefore, hypothesize that local officials with higher political incentives are more likely to faithfully enforce the UPSB policy. We test this hypothesis using the information of prefecture leaders, obtained by surveying the local government websites and through Baidu Baiken (Chinese version of Wikipedia). Higher promotion incentives are defined as the age of prefecture leaders (both party secretary (PS) and mayor) is below 57 (Axbard and Deng, 2024; He et al., 2020b; Nian, 2023), since the chance of promotion decreases significantly after this age. Table 3 shows that the policy effects are largely driven by local officials (both the party secretary and mayor) with higher promotion incentives, supporting our hypothesis that political incentives play a key role in this campaign-style environmental regulation.

Top-down accountability. The success of the UPSB policy also depends on how effectively upper-level governments monitor the performance of their subordinates (Axbard and Deng, 2024). We then

³⁶ Results are quantitatively the same if we instead use the unweighted average or use alternative weights, e.g., the number of first-time switchers that used in the estimation.

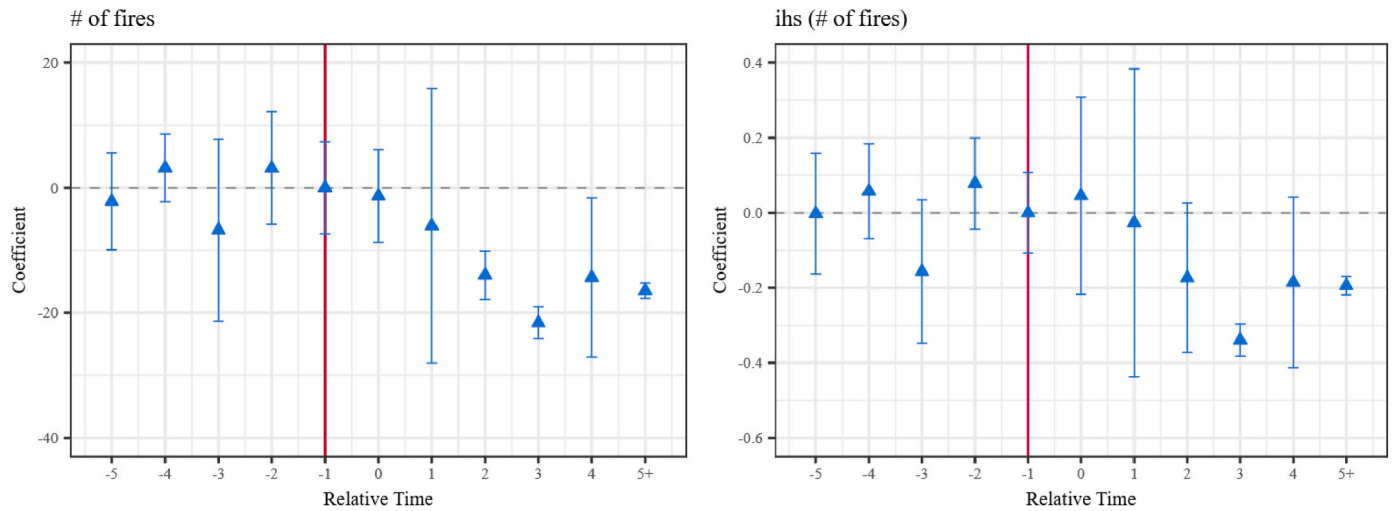


Fig. 5. DiD estimates of the UPSB policy on agricultural fires

Notes: This figure plots the coefficients from the DiD estimator. The left panel plots the UPSB policy effects on the absolute number while the right panel plots the effects on the IHS-transformed number of fires. The unit of observation is $10 \text{ km} \times 10 \text{ km}$ grid cells. The estimation exploits the `did_multipligt` command in Stata. Coefficient estimates are plotted with the 95% confidence interval. All regressions include geographic and weather controls. Bootstrap standard errors are clustered at the group (here, the group is defined at the province) by year level.

Table 3

The effects of political incentives.

Dep. Var.	(1)	(2)	(3)	(4)
	Number of Fires		IHS (# of Fires)	
<i>Post</i> × <i>Cropland</i> × <i>High Incentive (PS)</i>	−13.156** (5.620)		−0.308*** (0.082)	
<i>Post</i> × <i>Cropland</i> × <i>High Incentive (Mayor)</i>		−11.898** (5.592)		−0.167* (0.087)
<i>Post</i> × <i>Cropland</i>	−4.085 (5.253)	−6.406 (4.797)	0.012 (0.075)	−0.080 (0.068)
Observations	1,716,717	1,716,717	1,716,717	1,716,717
Grid FE	Yes	Yes	Yes	Yes
County by Year FE	Yes	Yes	Yes	Yes
Group by Year FE	Yes	Yes	Yes	Yes
Geo Controls	Yes	Yes	Yes	Yes
Weather Controls	Yes	Yes	Yes	Yes
Dep. Var. Mean	5.805	5.805	0.125	0.125
Adjusted R-squared	0.540	0.540	0.549	0.549

Notes: The unit of observation is $10 \text{ km} \times 10 \text{ km}$ grid cells. This table tests for the political incentive mechanism. The dependent variables are the number of fires and the IHS transformation of agricultural fires. *Post* is an indicator for years after the policy implementation. *Cropland* is the cropland share for each grid cell. *High Incentive* is a dummy indicator that equals one for city leaders (i.e., party secretaries (PS) and mayors) aged below 57. Geographic controls include slope, ruggedness, elevation, distance to the county border, county center, and provincial center; distance to rivers, and the number of rivers; distance to airports, distance to expressways, distance to railways, and distance to national roads and provincial roads. Weather controls include temperature, precipitation, humidity, sea level pressure, wind speed, and wind direction. Standard errors in parentheses are two-way clustered at the grid and county-year level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

investigate this by leveraging two sources of variation. First, we analyze whether the grid cells located upwind from their nearest air pollution monitoring station within the same prefecture face more stringent regulations.³⁷ The large-scale automated air pollution monitoring program has been staggered rollout since 2013, aiming to improve urban air

quality (Axbard and Deng, 2024; Greenstone et al., 2022; Xie and Yuan, 2023). According to Xie and Yuan (2023), firms located upwind from monitoring stations face stricter regulations. In practice, however, the monitoring stations can also detect air pollution originating from the straw-burning activities in rural areas (Guo, 2021; He et al., 2020a).³⁸ We therefore hypothesize that grids located upwind of monitoring stations experience more stringent top-down accountability. Results in Table 4, columns (1) and (3), confirm that these grids face more stringent regulations, implying that enhanced monitoring technology facilitates more effective environmental regulations.

Second, we consider the role of enhanced on-site supervision from the upper-level government. To discipline the performance of local agents, the superiors often adopt measures like random on-site inspections. Anecdotal evidence from Zhou (2022) suggests that expressways can promote impromptu inspections by enabling upper-level government officials to conduct inspections more flexibly and promptly. Results in columns (2) and (4) of Table 4 imply that our treatment effects are largely driven by grid cells that are closer to the expressway, aligning with our narratives.

We also rule out some competing mechanisms. For instance, if the UPSB policy intensified enforcement in no-burning zones (NBZs), this might explain the larger treatment effects observed near expressways, as fires that occur within the NBZs can result in severe accidents. To test this, in Appendix Table C1, we examine how the treatment effect varies across the distance to other NBZs (e.g., airports, railways, national and provincial roads). Contrary to the above hypothesis, we do not find any significant evidence suggesting that the stringency of the UPSB policy is stronger near these NBZs. Another related issue is the bottom-up mechanism in the top-down enforcement. This may arise if local officials themselves have the incentive to control the agricultural fires, even if it is costly. As documented by Dipoppa and Gulzar (2024), local officials may be more motivated to regulate agricultural fires if fires affect most of the home district. Similarly, local officials would be more motivated to regulate crop burnings near the government headquarters, as the burning may bear health costs to the officials. In Appendix Table C2, we examine whether the UPSB policy effects differ in locations

³⁷ Appendix Figure C1 gives an illustration of how we determine the upwind region, following the common practice in the literature (Axbard and Deng, 2024; He et al., 2020a; Rangel and Vogl, 2019; Xie and Yuan, 2023).

³⁸ Some provinces also strengthen the enforcement of the UPSB policy in the dominant upwind region, see for example <http://sthjt.gxzf.gov.cn/gxhd/myzz/W020200327288457747562.pdf>. Accessed at 2024-04-29.

Table 6
Farmers' responses to the UPSB policy.

Dep. Var.	(1)	(2)
	Fertilizer Usage (Standardized)	
<i>Post × Cropland</i>	0.033** (0.014)	0.030** (0.014)
Observations	1,650,006	1,649,574
Grid FE	Yes	Yes
County by Year FE	Yes	Yes
Group by Year FE	Yes	Yes
Geo Controls	No	Yes
Weather Controls	No	Yes
Dep. Var. Mean	2.923	2.923
Dep. Var. Standard Deviation	5.738	5.738
Adjusted R-squared	0.991	0.991

Notes: The unit of observation is 10 km × 10 km grid cells. This table investigates how fertilizer usage responds to the UPSB policy. The dependent variable is standardized Nitrogen fertilizer usage intensity (Yu et al., 2022). *Post* is an indicator for years after the policy implementation. *Cropland* is the cropland share for each grid cell. Geographic controls include slope, ruggedness, elevation, distance to the county border, county center, and provincial center; distance to rivers, and the number of rivers; distance to airports, distance to expressways, distance to railways, and distance to national roads and provincial roads. Weather controls include temperature, precipitation, humidity, sea level pressure, wind speed, and wind direction. Standard errors in parentheses are two-way clustered at the grid and county-year level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

the grid cell level. Instead, we resort to a coarser measure at the provincial level, which provides annual total pesticide usage as well as total fertilizer usage (by different categories) to complement our analyses.⁴¹ However, using more aggregated data has limitations, as it prevents us from capturing granular variations, and the results may be confounded by other unobserved heterogeneity. Thus, the results should be interpreted with caution. With this caveat in mind, we report our estimated coefficients in Appendix Table D1 and Table D2. In Appendix Table D1, columns (1) and (4), we regress the total fertilizer and pesticide usage on the treatment dummy, conditional on the province and year fixed effects, and treatment-specific linear trends to absorb potential differential trends between treatment and control provinces. In columns (2) and (5), we include a set of controls that may be correlated with agricultural production as well as fertilizer and pesticide usage. In columns (3) and (6), we weight our regressions by rural population a la Fletcher and Noghanibehambari (2024). Across different specifications, we find consistent and significant evidence that the UPSB policy unintentionally increases total fertilizer and pesticide usage.

In Appendix Table D2, we further examine fertilizer usage by different categories (i.e., nitrogen (N), phosphorus (P), and potassium (K)). One potential is that although nitrogen is essential for crop growth, the ash from straw burning contains mainly potassium. Thus, the increase in nitrogen fertilizer could be due to other unobserved negative shocks (although the increase in pests and disease risk may also incentivize higher nitrogen fertilizer use). But if we do not find any increase in potassium fertilizer, the concern would be exacerbated since potassium is a direct substitute for ash. Reassuringly, in Appendix Table D2, we find that both nitrogen and potassium fertilizer usage have increased after the UPSB policy, which lends further credence to our hypothesis that farmers indeed increased fertilizer usage in response to the policy.

Similar results are observed using household-level data, as reported in Appendix Table D3. We find strong evidence that household increased their fertilizer and pesticide usage per plot of land, and correspondingly, the expenditure spent on these factor inputs has significantly increased.

We provide further validation to the above results by examining

whether the UPSB policy leads to increased weeds and pests. To perform the test, we exploit data on the occurrence of major crop field diseases in China, which was conducted in the years 2013, 2017, and 2018 (covering the pre-policy and post-policy periods).⁴² The dataset was collected through field surveys on crops across several provinces, mainly investigating the occurrence of weeds for maize and wheat. It includes detailed survey locations, which allows us to geocode the data and merge it with our grid-level dataset. The dataset also includes information on the occurrence density and coverage intensity of weeds, which we exploit to measure the intensity of crop diseases. We present the results in Appendix Table D4, where we find that the UPSB policy significantly increases the occurrence density and coverage intensity of weeds on wheat, while the magnitude for maize is small and not significant at the conventional level. This lends additional credence to our assertion that the UPSB policy indeed increases the occurrence of weeds and pests, which leads farmers to use more fertilizers and pesticides as input adjustment.

Discussion. While the above findings provide strong evidence that banning straw burning and implementing straw returning increases fertilizer and pesticide use, there is contradictory evidence in the existing agronomic and crop science literature. For example, several studies suggest that returning crop residues to the soil (instead of burning) is associated with improvements in soil health that can aid crop growth and productivity (e.g., Fu et al., 2021), and burning itself is linked to long-term detrimental effects on soil quality (Amorim et al., 2021; Dutta et al., 2022). To reconcile our evidence with the existing literature, we provide two pieces of arguments.

First, we note that the relation between straw burning/returning and soil quality depends crucially on the local environment. Temperature, precipitation, and cropping traditions (e.g., double cropping, rotation cropping, and intercropping), among others, can have a significant impact on whether it is suitable for straw returning. In China, due to the prevalence of double cropping and crop rotations, the period available for straw decomposition and fermentation during crop intervals is extremely limited. Several studies also find consistent evidence that straw returning could increase the risk of root rot, pest infection, and other diseases (Zhu et al., 2014; Wang et al., 2020; Zhang et al., 2022). This also aligns with our evidence that weed increases following the implementation of the UPSB policy.

Second, even if straw returning is beneficial to soil quality, farmers' priori belief may also drive up fertilizer and pesticide application. If farmers believe that straw burning is conducive to soil quality, then banning straw burning can increase fertilizer use even if there is no actual decline in soil quality. This is exactly the dilemma proposed by (Oraby et al., 2025), which suggests that burning is a culturally embedded practice and is believed to enhance soil fertility.⁴³ Therefore, how farmers perceive the practice of burning is also an important driver of their behavior in responding to government regulation.

On the empirical front, we provide three pieces of evidence to complement the above arguments. First, we present the event study estimates of grid-level fertilizer application in Appendix Figure D1, which is informative about the dynamic responses of fertilizer usage. If, in the long run, straw returning is conducive to soil quality, then the observed increase in fertilizer usage only represents a short-term adjustment. Our evidence from Figure D1 suggests that the response is less likely to be a short-term adjustment, as we observe a significant and

⁴² The data is derived from the National Earth System Science Data Center, National Science & Technology Infrastructure of China (<http://www.geodata.cn>). See detailed description at <https://www.geodata.cn/data/datadetails.html?dataguid=188509254879198&docId=10613>. Accessed at 2024-09-02.

⁴³ Whether straw burning is conducive to soil quality remains a debate in the agronomic literature (Roper et al., 2021). Based on our interview with local cadres, the effect of straw burning on soil fertility also depends on the soil characteristics (e.g., soil moisture and soil type).

⁴¹ The data are drawn from the Provincial Statistical Yearbook.

persistent increase throughout post-treatment periods.

Second, we conduct two heterogeneity exercises to help better understand how the local environment affects farmers' fertilizer application decisions. As suggested by the meta-analysis of Zhang et al. (2022), temperature and precipitation act as important catalysts in the process of straw decomposition, and they show that the effect of straw returning on pest disease is more significant in regions with lower temperature and lower precipitation. Therefore, we should find a stronger effect of fertilizer responses to UPSB policy in these regions. We provide consistent evidence in Appendix Table D5, where, using a similar triple-difference specification as in Table 5, we show that the policy effect on chemical fertilizer is more pronounced in grids with lower temperature (column (1)) and lower precipitation (column (2)). Finally, we investigate the effect of UPSB policy on soil quality, which is measured by soil organic carbon content.⁴⁴ While being an imperfect proxy, this measure allows us to empirically investigate whether straw returning in China is conducive to soil quality. We provide the corresponding results in Appendix Table D6, where we find significant evidence that UPSB policy leads to substantial decreases in soil quality. Specifically, our estimates suggest a 0.055 SD decrease in soil quality, roughly a 9.6 percent change relative to the sample mean.⁴⁵

6.2. The effect on water pollution

In this subsection, we investigate how the unintended increase in fertilizer and pesticide usage due to the implementation of the UPSB policy can deteriorate the water quality. As there is no grid-level water pollution measure, we resort to data from water quality monitoring stations, which are widely used to examine the causes and consequences of water pollution (Fan and He, 2023; He et al., 2020b). A key challenge in exploiting monitoring station-level data is how to integrate it with our grid-level data. Theoretically, one could either aggregate grid-level data to the station level or disaggregate station-level data to the grid level. On the one hand, aggregating the grid-level data to the station level avoids the manipulation of the outcome variable (e.g., water quality) but requires additional assumptions on defining the treatment intensity for each station. Moreover, the relatively small sample size at the station level may invalidate the statistical inference, leading to insignificant results. And since we need to abstract the grid level information when doing the aggregation, we must rely on less information at the aggregate level to conduct further analyses (e.g., exploring the heterogeneity). On the other hand, however, disaggregating station-level data to the grid level involves the manipulation of the outcome variable, which also requires strong assumptions, and the outcomes are likely to be measured with error. Nevertheless, the abundance of grid-level information allows us to explore potential mechanisms and provides more detailed analyses. As both approaches have their own pros and cons, we adopt both methods to ensure that our results are robust.

6.2.1. Aggregate grid-level data to the station level

Baseline estimates. First, we aggregate our grid-level data to the station level by mapping the two datasets and assigning each grid to its nearest monitoring station. We filter grids within a 100 km radius of their nearest station, as fertilizer increases in more remote grids should

have less impact on water pollution detected by the monitoring station. We then calculate the weighted average of cropland shares for each station (as the measure of treatment intensity), with the distance to the station serving as the weight. Controls are collapsed to the station level in a similar fashion. We then run the following regression to examine the effects of UPSB policy on water pollution.

$$y_{sgpt} = \beta \text{Post}_{pt} \times \text{Cropland}_s + X_s \times T_t + Z_{st}\theta + \delta_s + \eta_{pt} + \zeta_{gt} + \epsilon_{icgpt} \quad (4)$$

Where y_{sgpt} represents the station-year level outcome. We use water quality grades (a categorical variable ranging from 1 to 6, with the higher the value, the worse the quality), COD, and $\text{NH}_3\text{-N}$ to measure the degree of water pollution. We standardize the water quality grades to a mean of 0 and a standard deviation of 1, following Perez-Truglia (2020) by using the Probit-OLS method to assign values. Cropland_s measure treatment intensity for station s , aggregated from the grid level. Geographic controls and weather controls are similarly defined at the station level. We also include station-level fixed effects, δ_s , province-year fixed effects, η_{pt} , and treatment group by year fixed effects, ζ_{gt} , with identical definitions to the previous specifications. Standard errors in parentheses are two-way clustered at the station level and the province-year level.

Table 7 reports the results based on equation (4). In columns (1), (3), and (5), we find strong evidence that the UPSB policy resulted in worsened water quality grades and increased COD and $\text{NH}_3\text{-N}$ pollution. With an average cropland share of about 0.25 surrounding the station, our estimates suggest that the UPSB policy generally leads to a deterioration in water quality grades by 0.168 SD, and increases in COD and $\text{NH}_3\text{-N}$ by 0.1 and 0.05 units (which translate into an increase of 2.3% and 9.1% relative to the mean), respectively. In columns (2), (4), and (6), we test the robustness of using an alternative bandwidth for calculating the treatment intensity within a 50 km radius of each station. The results remain unaffected. This is in line with our interpretation that the water pollution detected by the monitoring stations is largely caused by grids near the station.

We examine the identifying assumption underlying the above specification using an event study estimation. The results are reported in Fig. 6. We find no significant pre-trends for all three outcome variables, suggesting that the parallel trends assumption holds plausibly. After the implementation of the UPSB policy, we find that water pollution significantly increases, with stronger effects in subsequent years. This alleviates concerns of omitted variable bias and reverse causality, as we only detected significant effects in periods after policy implementation, and in stations with higher treatment intensity. Any other confounding factors must simultaneously mimic the temporary variation in treatment timing and cross-sectional variation in treatment intensity, which, we believe, is less plausible. In addition, the results are also robust if we alternatively use the DiD₁ estimator proposed by de Chaisemartin and D'Haultfœuille (2024), which helps to aid the potential negative weighting issues (shown in Appendix Figure D2).

Robustness. We conduct several robustness checks to ensure that our results are not driven by other confounding factors. First, we run a falsification test that randomly assigns a cropland share to each station, and interacts the placebo shares with the UPSB policy dummy. If the increased water pollution is driven by reduced agricultural fires (i.e., the pollution substitution effects), we should find no effects in these placebo specifications. We conduct the falsification test for 500 times and plot the kernel density of estimated coefficients in Appendix Figure D3, where we find that the placebo coefficients are all centered at zero, and the magnitudes from the true specification are way larger than these placebo estimates.

Second, we control for other possible factors that may affect water pollution. A leading confounder is the selective water pollution regulations. As documented in the literature (Cai et al., 2016; Lipscomb and Mobarak, 2017), water pollution is significantly rampant at administrative boundaries due to its negative externality, which leads to a

⁴⁴ The data is provided by the National Tibetan Plateau Data Center (TPDC). TPDC offers the 1 km grid-level soil information from 2010 to 2018, which was created by Liu and Zhang (2021). We follow the same procedure described in the Data section to process the data and aggregate them to our 10 km grid level. See more detailed descriptions of the dataset at <https://www.tpdc.ac.cn/zh-hans/data/e1ccd22c-348f-41a2-ab46-dd1a8ac0c955/>.

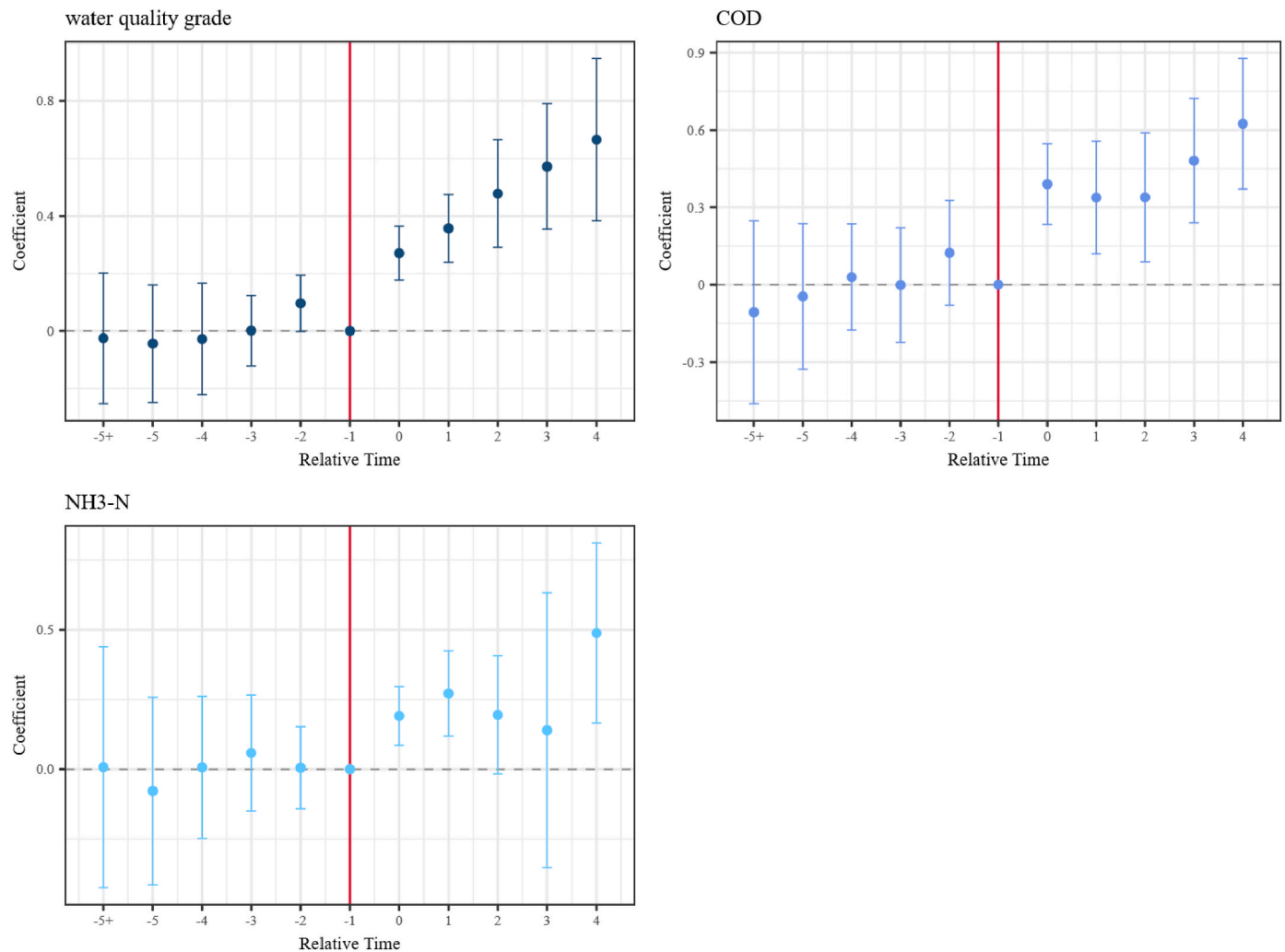
⁴⁵ One potential concern with our soil quality results is that the increase in fertilizer and pesticide application may also contribute to the decline in soil quality. To ensure this is not the case, we control for fertilizer usage in columns (3) and (4) of Table D6; the results remain robust.

Table 7

The effects of UPSB policy on water pollution (station level).

Dep. Var.	(1)	(2)	(3)	(4)	(5)	(6)
	100 KM	50 KM	100 KM	50 KM	100 KM	50 KM
	Water Quality Grade (Standardized)		COD		NH3-N	
<i>Post</i> × <i>Cropland</i>	0.673*** (0.184)	0.562*** (0.141)	0.400** (0.188)	0.462** (0.175)	0.208** (0.077)	0.209*** (0.067)
Observations	1386	1386	1511	1511	1513	1513
Station FE	Yes	Yes	Yes	Yes	Yes	Yes
Province by Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Group by Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Geo Controls	Yes	Yes	Yes	Yes	Yes	Yes
Weather Controls	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Var. Mean	0	0	4.272	4.272	0.547	0.547
Adjusted R-squared	0.794	0.793	0.852	0.852	0.790	0.790

Notes: The unit of observation is at the station-year level. This table presents the results of the UPSB policy on water pollution. The dependent variables are Water Quality Grade (standardized as la Perez-Truglia (2020)), chemical oxygen demand (COD), and ammonia nitrogen (NH₃-N). *Post* is an indicator for years after the policy implementation. *Cropland* is the cropland share for each station, aggregated from the grid level. Geographic controls include slope, ruggedness, elevation, distance to the county border, county center, and provincial center; distance to rivers, and the number of rivers; distance to airports, distance to expressways, distance to railways, and distance to national roads and provincial roads. Weather controls include temperature, precipitation, humidity, sea level pressure, wind speed, and wind direction. Standard errors in parentheses are two-way clustered at the station level and the province-year level. * denotes significance at the 10% level. ** denotes significance at the 5% level. *** denotes significance at the 1% level.

**Fig. 6.** Event study estimates of the UPSB policy on water pollution

Notes: This figure plots the estimated event-study coefficients of the impact of the UPSB policy on water pollution. The upper left panel plots the effects on standardized water quality grade, the upper right panel plots the effects on COD, and the lower left panel plots the effects on NH₃-N. The unit of observation is at the station-year level. Regression specification is presented in Equation (4). Coefficient estimates are plotted with the 95% confidence interval. All regressions include geographic and weather controls. Standard errors in parentheses are two-way clustered at the station level and the province-year level.

“polluting thy neighbor” effect where local officials may strategically reduce regulation efforts at the boundaries. Another potential confounder is emissions from water-polluting firms. As the readings from monitoring stations cannot distinguish whether the pollution is from industrial pollution or agricultural pollution, deteriorating water quality can also be attributed to increased firm emissions. To ensure that our estimated effects are not driven by the selective regulation, we generate a dummy variable indicating whether a monitoring station is located at the provincial border and interact it with a full set of year fixed effects to control for the effects of potential selective regulation. Similarly, to ensure that the effects are not confounded by firm emissions, we calculate the number of water-polluting firms within a 100 km radius of each station and interact with a full set of year fixed effects.⁴⁶ Reassuringly, our estimated effects on water pollution remain significant and quantitatively unchanged (Appendix Table D7).

Third, to ensure that our results are not driven by other local contemporaneous programs, we additionally examine the robustness of our results to the inclusion of watershed-by-year trends or watershed-by-year fixed effects, which allows us to control for local environmental regulations specific to certain watersheds.⁴⁷ The corresponding results are reported in Appendix Table D8, where we do not find large changes in our estimated coefficients. Lastly, to ensure that our results are not contaminated by upstream water station readings, we exclude immediately adjacent and upstream monitoring stations that are located in the same watershed. The results from Appendix Table D9 suggest that our estimated effects are less subject to the intercorrelation between upstream-downstream water stations.

Heterogeneity. Similar to the heterogeneity discussion in Section 5.2, we can explore how political incentives and enforcement costs (discussed next) affect the magnitude of water pollution. Intuitively, as we’ve shown that the effect of UPSB policy on reducing agricultural fires is more significant when local officials have higher promotion incentives, then we should expect a symmetric effect on water pollution, i. e., strong political incentives lead to more significant increases in water pollution. In Table 8, we replicate the specification from Table 3 and examine whether the increase in water pollution following the UPSB policy is largely driven by promotion incentives of local officials. The corresponding results confirm our argument, where we do find that the effect of UPSB policy on water pollution is more significant if local officials, especially the mayor, have higher incentives. Such a symmetric structure between the reduction in agricultural fire/air pollution and the increase in water pollution lends further confidence to our identification of the pollution substitution effect.

6.2.2. Disaggregate station-level variables to the grid level

Baseline estimates. To cross-validate our findings and to provide more evidence on how the UPSB policy leads to unintended water pollution, we adopt an alternative strategy by disaggregating station-level variables into the grid level. To do so, we need to assign station-level water pollution readings to grid observations. Since water pollution is measured at specific monitoring stations, only grids that are closer to monitoring stations can exert effects on observed water pollution. We thus restrict our grid-level sample to grids within a 100 km radius of the nearest monitoring station. To avoid any manipulation of the raw water pollution data, we assign the exact monitoring readings to those grids as the outcome variable at the grid level.

We rerun our baseline regression using the inverse of the distance to

the nearest station as weights, since fertilizer usage in grids closer to the monitoring station may contribute more significantly to water pollution. As outcomes are the same across grids that belong to the same station, we therefore control for the station fixed effects and cluster the standard errors at the station level. Reassuringly, the results reported in Table 9 again confirm that all three water pollution measures increased after the implementation of the UPSB policy. Moreover, the estimated coefficients from Table 9 are similar to those obtained in Table 7, which lends additional credence to the validity of our estimates. Specifically, our estimation from Table 9 suggests that, when evaluated at the mean cropland share (0.192), the implementation of the UPSB policy leads to a degradation of water quality by 0.129 SD, and increases COD and NH₃-N by 0.095 and 0.025 units (which translate into an increase of 2.2% and 3.3% relative to the mean), respectively.

We also examine the identification assumption for the grid-level sample. In Appendix Figure D4 and Figure D5, we report the event study estimates using both the fixed effects estimator and the DiD estimator. Both estimators show no significant pre-trends, which again provides valid support for the parallel trend assumption and ensures that our results are not driven by the potential negative weighting issues.

Robustness and heterogeneity. The grid level specification allows us to conduct several heterogeneous exercises and robustness checks to verify that we are accurately estimating the effects of UPSB policy on water pollution. First, similar to our previous finding that water pollution increases more if local officials have higher promotion incentives, the effects of UPSB policy on water pollution should be weaker if the enforcement cost is relatively higher. We examine this heterogeneous effect by replicating the results from Table 5, with the dependent variable replaced with corresponding water pollution measures. Table 10 reports the results, using temperature and terrain ruggedness as measures of enforcement cost. Symmetrically, we find the effects on water pollution are lower if the temperature is lower and the terrain is more rugged. This aligns with the results from Table 5, which state that the enforcement cost is higher if the temperature is lower or the terrain is more rugged.

Second, since worsened water pollution is primarily driven by the overuse of fertilizers and pesticides in our setting, stronger effects should be observed in grids with higher fertilizer usage. To examine the mediating effects of fertilizer usage, we analogously estimate an augmented model by interacting the fertilizer variable with our key independent variable. The results in Appendix Table D10 show that the estimated effects on water pollution are stronger in grids with more intensive fertilizer usage.

Third, as another indirect validation, we expect stronger effects on water pollution in grids with higher precipitation. Increased precipitation leads to more surface runoff, which dissolves the fertilizers and pesticides applied on the cropland and carries them to nearby water bodies, thus exacerbating the negative externalities of fertilizer and pesticide usage. If increased water pollution is not driven by fertilizers and pesticide overuse (e.g., stemming from industrial pollution), we should observe no heterogeneous effects, as the industrial activities should be less relevant to these conditions. Appendix Table D11 confirms this argument, where we find larger and more salient effects on water pollution in grids with higher precipitation, reinforcing the causal link between capturing the effects of UPSB policy on water pollution.

Fourth, in our baseline specification, we use cropland shares of grids within a 100 km radius of their closest monitoring station to construct the treatment intensity. Since only upstream fertilization activities can be detected by the monitoring station, we should observe effects solely from grids that are located upstream of the monitoring station. We use a DEM raster to predict river flow direction and determine upstream grid direction. Appendix Table D12 reports the results from separate regressions for upstream and downstream subsamples. Reassuringly, we find consistent evidence that the effects on water pollution are entirely driven by the upstream subsample, providing further support to our findings.

⁴⁶ The data on polluting firms are drawn from the Environmental Survey and Reporting Database (ESRD), which is compiled by the Ministry of Ecology and Environment (MEP), that records all heavy polluters that collectively contribute to 85% of total emissions (He et al., 2020b). We follow the guidelines of MEP to select water-polluting firms.

⁴⁷ Examples include the recent transboundary ecological compensation programs launched in several Chinese provinces (Wang et al., 2025).

UPSB policy would lead to a decrease of 10.4 agricultural fires (column (3)). Combined with the estimates from He et al. (2020a), which suggest that 10 additional crop fires lead to a $4.79 \mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$.⁵⁰ This implies that the UPSB policy would decrease $\text{PM}_{2.5}$ by $4.98 \mu\text{g}/\text{m}^3$.⁵¹ We then relate the estimated effects on air pollution with studies that investigate the mortality effects of pollution to back out the health benefit of the UPSB policy. In particular, estimates from Fan et al. (2020) suggest that a 10-point increase in $\text{PM}_{2.5}$ would increase the mortality rate by 2.5%.⁵² Given that the mortality rate documented by the Chinese Center for Disease Control and Prevention's (CCDC) Disease Surveillance Points (DSP) system is 131.76 per 100,000 people (Fan et al., 2020),⁵³ this implies that the UPSB policy could save 1.64 lives per 100,000 population through improved air quality.

On the other hand, consider the policy effects on water pollution. Our estimates from Table 7 reveal that the implementation of the UPSB policy would lead to an increase in water quality grade by 0.56 standard deviation (column (2)), which approximately translates into 0.46 unit deterioration of water quality. Combined with the estimates from Ebenstein (2012), which find that an escalation of water quality by a single grade increases the digestive cancer death rate by 9.7%. Given the average mortality rate of digestive cancer death is 26 per 100,000 people (Ebenstein, 2012), these figures imply that the UPSB policy could lead to 1.16 additional deaths per 100,000 population through degraded water quality. Altogether, the above statistics suggest that the UPSB policy, when evaluated as a relative change in treatment intensities, could only decrease 0.48 additional deaths per 100,000 population.⁵⁴ Through comparison, we find that much of the health benefit (approximately 70%) from improved air quality is offset by the health cost from degraded water quality.

Overall change. The above estimates only reveal the relative change in health benefit due to changes in treatment intensity. To evaluate the overall policy impacts, we now rely on our estimates from equation (1), which directly compares pollution outcomes between treated and control provinces, before and after the policy implementation. The estimated results on agricultural fires are reported in Appendix Table A5. We additionally report the effects on water pollution estimated from equation (1) in Appendix Table D14. Following the same procedure as above, we find that the implementation of the UPSB policy decreases 5.83 number of agricultural fires, which turns into a reduction in $\text{PM}_{2.5}$ by $2.79 \mu\text{g}/\text{m}^3$ and approximately saves 0.92 lives per 100,000 population. Turning to the estimates of water pollution, our results imply that the implementation of the UPSB policy increases water quality grade by 0.12 SD, which translates into 0.15 unit degradation in water quality grade, and results in 0.38 additional deaths per 100,000 population. Overall, the UPSB policy saves 0.54 additional deaths per 100,000 population, which is similar to our previous calculation, that the

unintended increase in water pollution offset a substantial amount of health benefit due to improved air quality (approximately 41.3%). Given the total population in all treated provinces is approximately 0.71 billion, the UPSB policy could save 3834 lives in net.

Using the value of statistical life (VSL), we can further monetize the health benefit of the UPSB policy. Specifically, Fan et al. (2020) adjust the VSL from Qin et al. (2013) and estimate that the VSL is around 7.46 million CNY or 1.15 million USD in 2015. We adopt their calculation and similarly discount it by 30% since the elderly are more susceptible to pollution exposure (Fan et al., 2020). This yields a net benefit of 20.02 billion CNY or 3.08 billion USD. Decomposing the net health benefit into gain in air quality and loss in water pollution, we find that the gain from improved air quality is 34.11 billion CNY or 5.25 billion USD. In comparison, the loss from degraded water pollution is 14.09 billion CNY or 2.17 billion USD.

If we are willing to impose several strong assumptions, we can take one step further to consider the enforcement cost of the UPSB policy (including the administration costs of regulating straw burning and the private agricultural production costs, e.g., input adjustments and yield losses). This allows us to compare the policy cost with its net health benefit, and to determine whether the UPSB policy is cost-effective. However, due to space limitations and to avoid potential distraction of our empirical findings, we regrettably relegate this part of the discussion to Appendix E for interested readers. In that Appendix, we discuss in further detail how to estimate the enforcement cost, especially the administration cost, of the UPSB policy, and how we can combine our causal estimates with the Marginal Value Public Funds (MVPF) approach and other estimates of willingness to pay (WTP) parameters to infer the welfare impact of the UPSB policy (Finkelstein and Hendren, 2020).

8. Conclusion

This paper investigates the effectiveness and pollution substitution effects of a command-and-control policy in China. We use the case of a recent straw-burning ban, which aims to regulate air pollution. We find that the straw burning ban significantly decreases agricultural fires and related air pollution. The political incentives, along with the top-down accountability associated with implementing straw-burning bans, are the driving forces. Heterogeneous analysis shows that the reduction of agricultural fires is weaker in grids with lower temperatures and in rugged terrain areas, where enforcement costs and monitoring costs are higher. We also find that the reduction effects are less significant in grids with high wind speeds.

However, we find that the top-down straw-burning regulation also leads to unintended consequences of increased water pollution. This result is driven by farmers' input adjustment in their use of chemical fertilizers and pesticides. Before the regulation, straw burning would naturally fertilize the soil and help control pests. With the ban in place, farmers have compensated by increasing their application of chemical inputs, which has, in turn, contributed to water pollution. This finding highlights that overlooking how farmers might strategically respond to the regulation can result in an unintended pollution substitution effect.

CRedit authorship contribution statement

Hai Hong: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization, Writing – review & editing. **Kevin Chen:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Conflict of interest

The authors declare no competing interests.

⁵⁰ See column (3) of Table 2 in their paper.

⁵¹ We do not use our estimated coefficients on air pollution, as it may be biased due to the strategic behavior of local officials. As the gain from reduced air pollution in regulating straw burnings may incentivize local officials to relax the regulation on other air pollution-related manufacturing firms, which would lead to an underestimate of the policy effects on air pollution.

⁵² We do not use the estimate from He et al. (2020a) since they do not provide the average mortality rate, but their estimates on the effects of air pollution on mortality are similar to the ones obtained by Fan et al. (2020).

⁵³ The measure of mortality rate in Fan et al. (2020) is at the monthly level; we multiply it by 12 to roughly approximate the annual mortality rate. This number aligns with estimates from other reports, see, for example, <https://rs.yiigle.com/CN112338202201/1349882.htm>. Accessed at 2024-12-19.

⁵⁴ We note, however, that these results should be interpreted with great caution as the estimated net health benefit crucially hinges on the mortality rate, which may change over time. Although both Ebenstein (2012) and Fan et al. (2020) use data from DSP, their research periods differ: Ebenstein (2012) studies mortality rates during 1991–2000, while Fan et al. (2020) conduct their analysis during 2014 and 2015.

Abstract

The widespread prevalence of unexploded landmines in many countries leads to thousands of deaths and injuries annually. In addition, landmines may impede the flow of goods and people; this not only restricts trade and labor flows, but also hinders access to schools and medical care. Moreover, landmines can block the productive use of contaminated land. We study the clearance of more than 15,000 hazardous areas in Afghanistan carried out since 1992, obtaining precise geographic boundaries for these areas. We identify a window during which a policy shift in clearance targeting created plausibly exogenous variation in the timing of clearance, which we study using a two-way fixed effects panel design. The clearance of hazardous areas leads to changes in land use observed using multispectral, moderate-resolution Landsat-series imagery, as well as to increases in economic activity reflected in nighttime lights data. Our precise clearance data and satellite imagery allows us to observe the shifts occurring even in the small towns and villages that comprise most of our sample. We also find reductions in conflict risks due to the clearance of hazardous areas. Landmine action efforts thus appear to have substantial economic effects even in areas with ongoing conflict risks.

Author Information

Christian Baehr

Princeton University

Ariel BenYishay

AidData, William & Mary

Rachel Sayers

AidData, William & Mary

Kunwar Singh

AidData, William & Mary

Madeleine Walker

AidData, William & Mary

The views expressed in AidData Working Papers are those of the authors and should not be attributed to AidData or funders of AidData's work, nor do they necessarily reflect the views of any of the many institutions or individuals acknowledged in the AidData Working Paper Series.

Acknowledgments

The authors are grateful to Anne-Li Naucler, Tom Gillhespy, and George Bowles for facilitating access to the mine clearance geospatial data and providing excellent comments, as well as to seminar participants at VCU, William & Mary, and the Center for Environmental Economics - Montpellier.

1 Introduction

The burgeoning geospatial data on conflict events has powered a new spate of research over the past decade, uncovering insights on the interplay between active conflicts and economic development (Sundberg and Melander, 2013; Fjelde and Hultman, 2014; Michalopoulos and Papaioannou, 2016; Uexkull et al., 2016; Wagner et al., 2018). Yet even after active conflict subsides, unexploded ordinances like landmines remain major dangers throughout much of the world. Because landmines cost relatively little to produce (typically \$3-\$30), they have been heavily used throughout the developing world, and more than 100 million landmines remained buried across 60 countries as of 2021 (ICBL, 2021). The detonation of unexploded landmines leads to thousands of deaths and injuries annually (Frost et al., 2017). In addition to being public health threats, landmines may also severely limit economic development. When located near paths or roads, they can impede the flow of goods and people, restricting not only trade and labor flows, but also access to schools and medical care. Moreover, landmines can restrict the productive use of contaminated and surrounding land, preventing fields from being sown and commercial activities from being expanded.

Yet, because precise geospatial data on landmines has been very limited, relatively little is known about the extent of the limitations they impose on development, or, conversely, about the gains in economic development afforded by landmine clearance. To date, only four studies have examined these development impacts. Three of these studies use cross-sectional instrumental variable models to compare populations based on their proximity to hazardous areas, finding worse child nutrition, greater poverty, and lower educational attainment in higher risk areas (Arcand, Rodella-Boitreaud and Rieger, 2015; Merrouche, 2008, 2011). More recently, Chiovelli, Michalopoulos and Papaioannou (2018) obtain data on both the location and timing of clearance activities in Mozambique, allowing them to adopt a panel design. These authors find large benefits from clearance of transport corridors and major hubs, but relatively limited effects in rural areas. This may be in part because they rely on nighttime lights emissions as their primary outcome measurement, and the lights data may not capture meaningful changes in unelectrified areas.

While these studies offer promising findings, major gaps in our knowledge remain. First, these studies focus on post-conflict settings, but many landmine clearance efforts take place in areas with ongoing conflict. Are the findings from post-conflict clearance representative of the gains from clearance when ongoing conflict risks might dampen broader economic changes? This may be particularly challenging to assess because the timing of clearance may depend in part on the ebb and flow of conflict, potentially confounding the roles of clearance and conflict dynamics. Secondly, although landmine clearance enables repurposing land, none of the aforementioned studies have examined land use as an outcome. Finally, there is still no clear evidence on whether landmine clearance leads to economic development in both rural and urban areas.

The latter question is critical to current policy debates, which focus on how to target the relatively limited landmine clearance efforts on an efficient and equitable basis. Landmine clearance organizations receive a total of \$650M annually worldwide, but with removal costs often exceeding \$1000 per mine, recent rates of removal have only reached a maximum of 200,000 mines annually (ICBL, 2021). Even ignoring the >2M new mines planted annually, full worldwide clearance would take >1,000 years at this rate. These constraints lead to fre-

quent targeting and prioritization debates, many of them centered on the relative importance of clearance in rural or agricultural land.

We provide several key contributions. First, we obtain a comprehensive, long-term, temporally and spatially precise dataset on hazardous area clearance in Afghanistan. Our spatial data includes the boundaries of each hazardous area, recorded during the actual clearance activities by each field team. This affords us greater statistical precision than previous studies, which relied on aggregated administrative units or geographically perturbed survey cluster locations. We combine the clearance data not only with nighttime lights data, but also with much finer land use changes in and around the cleared sites derived from multispectral (MS) satellite imagery. Our high-quality landmine clearance boundary data and use of MS imagery provides us with improved ability to detect land change over time in rural as well as urban and peri-urban areas. Unlike [Chiovelli, Michalopoulos and Papaioannou \(2018\)](#), we find some of the largest effects occur outside of urban areas. In fact, in villages, we observe building construction and expansion near newly cleared areas. At the same time, we do not find increased conversion of cleared land to farming, likely because water and fertilizer availability are the most constraining factors to agricultural production.

Moreover, by also incorporating time-varying conflict conditions, we show that landmine clearance can actually reduce conflict risks. These conflict reductions appear independent of economic improvements due to clearance, showing much more nuanced impacts from landmine clearance than previously understood.

Our paper proceeds as follows: in Section 2, we lay out the context of landmine clearance efforts in Afghanistan over the past two decades. In Section 3, we describe the geospatial data we obtain on the clearance activities, as well as the satellite and survey data we use. Section 4 provides our empirical strategy for identifying the impacts of clearance, and Section 5 lays out our results, each separated into subsections on nighttime lights (NL), land use, and conflict. We offer conclusions in Section 6.

2 Landmine Deployment and Clearance in Afghanistan

2.1 Overall context

The extent of landmine contamination in Afghanistan—as well as the scale and longevity of Afghanistan’s demining program—present a uniquely important context for understanding the impacts of demining on economic development in the face of ongoing conflict risks.

The Soviet Army was the first to plant landmines in Afghanistan during their occupation of the country from 1978-1989. In the four decades of war that followed, most major players, including the Taliban, Mujahideens, and many tribal groups, have used landmines in a variety of settings ([Chawla, 2011](#)).¹ Since 1978, over 40,000 casualties have resulted from landmines and other unexploded ordinances ([UNMAS, 2021](#)). Every province of Afghanistan has been contaminated with landmines, and despite the major progress in mine clearance that we detail below, nearly every province continued to have landmine contamination in 2021 (affecting over 1,500 communities in 253 districts) ([UNMAS, 2021](#)).

¹The United States is known to have used only one landmine during its presence in Afghanistan, in a Special Operations mission in 2002 ([Crowley and Ismay, 2022](#)).

While many landmines remain buried, Afghanistan has nonetheless made major progress in landmine clearance over the past 30 years. Between 1989 and 2021, 81.3% of known hazardous areas in the country were cleared or determined non-hazardous ([UNMAS, 2022](#)). Demining initiatives began when the United Nations established the Mine Action Program of Afghanistan (MAPA) in 1989. Between 1989 and 2008, the UN and a consortium of international NGOs managed mine clearance operations in the country. Landmine clearance in this period was restricted by a lack of strategic planning and the destruction of demining equipment in the conflict between the US-backed Northern Alliance and the Taliban in the wake of the 2001 World Trade Center attacks. Before 2008, at most 40 square km of minefields were ever cleared in a single year. In 2010, the number had increased to 65 square km. ([Paterson, 2012](#)).

Afghanistan gained official authority over national landmine clearance operations in January 2008, when mine action management transitioned to the Department of Mine Clearance (DMAC) in the Afghanistan National Management Authority (ANDMA), under the President's office. Concurrent with this transfer of oversight, MAPA went through a number of decentralizing reforms. Local staff were prioritized over international staff, and an operations reform restructured demining units into smaller units ([Paterson, 2012](#)). MAPA also encouraged local and international NGOs to adopt a cluster approach to clearance tasks. The new approach emphasized clearing multiple hazard sites in one area, instead of just high-priority tasks, intended to utilize equipment and personnel more efficiently. Operators also increased security by strengthening their engagement and local relationships ([Monitor, 2009](#)).

These reforms led to major changes in the overall rate of mine clearance, as well as in the distribution of which sites were cleared. Between 1989 and 2007, a total of 3,084 hazardous areas were cleared across the entire country (an average of approximately 171 each year), with these areas concentrated around provincial capitals, towns, and main transportation corridors. Between 2008 and 2020, 17,648 hazardous areas were cleared, with these areas spanning a much broader set of locations.

The clearance of mines has taken place despite ebbs and flows of conflict risks, potentially dampening the extent to which landmine clearance can benefit economic growth. Taliban attacks on civilians rose steadily in the years leading up to the US withdrawal from Afghanistan, increasing to 10,469 in 2021 ([SIGAR, 2021](#)). Despite high conflict levels, 1,470 mines were cleared per year on average between 2008 and 2020. Continuing conflict between the Taliban and resistance groups will likely put civilians at high risk of experiencing violence into the future ([ACLED, 2022](#)).

2.2 Mine Clearance Data

We obtain the Mine Action Geo Data from DMAC, corresponding to all hazardous area clearance activities carried out throughout the country between 1992 and 2020. The spatial data include the polygon boundaries of each confirmed or suspected hazardous area, recorded during the survey and clearance conducted by each ground team, as well as the precise dates of these activities. There are 20,749 areas reflected in this data, of which more than 85% (17,913) were cleared since 2008.

For illustration, Figure 1 maps the precise boundaries of cleared hazardous polygons in

4 Empirical Strategy

To identify the impacts of clearance, we utilize a panel difference-in-difference approach with two-way fixed effects (FEs). This approach capitalizes on the staggered nature of clearance activities, as well as the geospatial variation in hazardous areas being cleared. In this approach, hazardous areas are part of both the control and the treatment group, depending on the time of observation relative to clearance. Depending on the outcome, we utilize either grid cells that intersect with hazardous areas or the full hazardous area polygons as the spatial units of analysis.

4.1 Nighttime lights

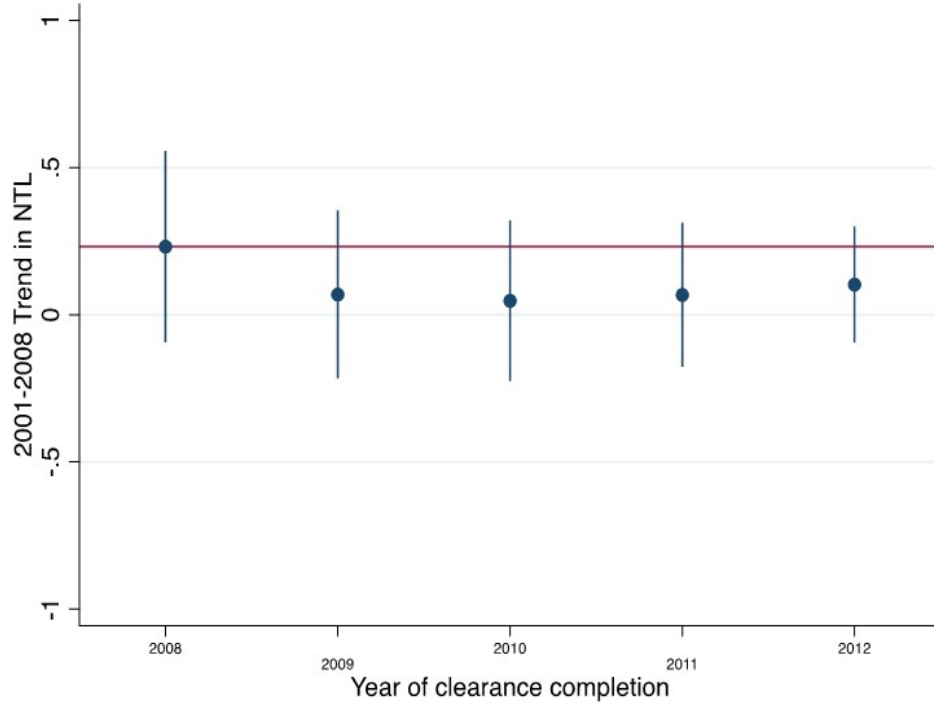
We estimate the effect of a grid cell being entirely cleared on NL using ordinary least squares (OLS) regressions with two-way FEs at the grid cell and province-by-year levels. The estimating equation for this analysis is:

$$NL_{jpt} = \alpha + \beta Cleared_{jpt} + \delta_j + \delta_{pt} + \epsilon_{jpt}$$

where NL_{jpt} represents the intensity of nighttime light emission from grid cell j in province p and year t . $Cleared_{jpt}$ is a binary variable taking a value of 1 if grid cell j is entirely cleared of landmines by year t , and 0 otherwise. δ_j is a vector of grid-cell fixed effects, which capture time-invariant local characteristics. δ_{pt} is a vector of year-by-province fixed effects, which capture geographically invariant characteristics within each year-by-province combination. We use two-way clustering of standard errors at the district and year level following [Cameron, Gelbach and Miller \(2011\)](#). Because we rely only on the within-grid cell variation in timing of clearance, we limit our sample to only grid cells cleared between 1992 and 2013 (i.e., we exclude cells cleared after 2013, when our NL data series ends).

This empirical strategy relies on the parallel trends assumption: individual grid-cells cleared earlier in our analysis window would have experienced similar trends to the not-yet-treated grid-cells had they not been cleared. As our FEs adjust for province-by-year-specific means, such differential trends would have to exist for earlier-cleared grid-cells compared to not-yet-cleared cells within the same province. To validate this assumption, we first use the long time-series of NL data preceding treatment to confirm that there were not differential pre-trends for earlier-cleared grid-cells. We estimate a linear model over the 2001-2008 time window with continuous time trends specific to each distinct year of clearance completion (along with the grid-cell and province-by-year fixed effects used in our main specification). The results, plotted in Figure 5, show little evidence of systematically different pre-trends based on the year of clearance completion.

Figure 5: Pre-trends in Nighttime Lights



We further validate our empirical strategy using recently developed approaches accounting for potential biases in panel models with two-way FEs. These biases arise when the treatment effects vary over time or over other characteristics, and can cause the standard two-way FE models to diverge substantially from the intended interpretation as the average effects comparing already treated to not-yet-treated units. We adopt the approaches of both [Chaisemartin and d’Haultfoeuille \(2020\)](#) and [Callaway and Sant’Anna \(2021\)](#) to account for these biases. The event study results produced by these approaches (shown in Section 5) also confirm the lack of differential pre-trends, giving further credence to our identifying assumption.

4.2 Land use

To estimate the impact of clearance on land use, we utilize our data on LULC categories derived from RF algorithms based on MS for five-year time intervals from 1995 to 2020. We generate this data for 30 m x 30 m grid-cells. In order to assign land use classifications to entire hazardous areas, we calculate the percentage of each hazardous area devoted to distinct land use categories. As such, the unit of observation is the hazardous-area-polygon-by-year level.

The estimating equation for this analysis is:

$$LandUse_{uipt} = \alpha_u + \beta_u Cleared_{ipt} + \delta_{ui} + \delta_{upt} + \epsilon_{uipt}$$

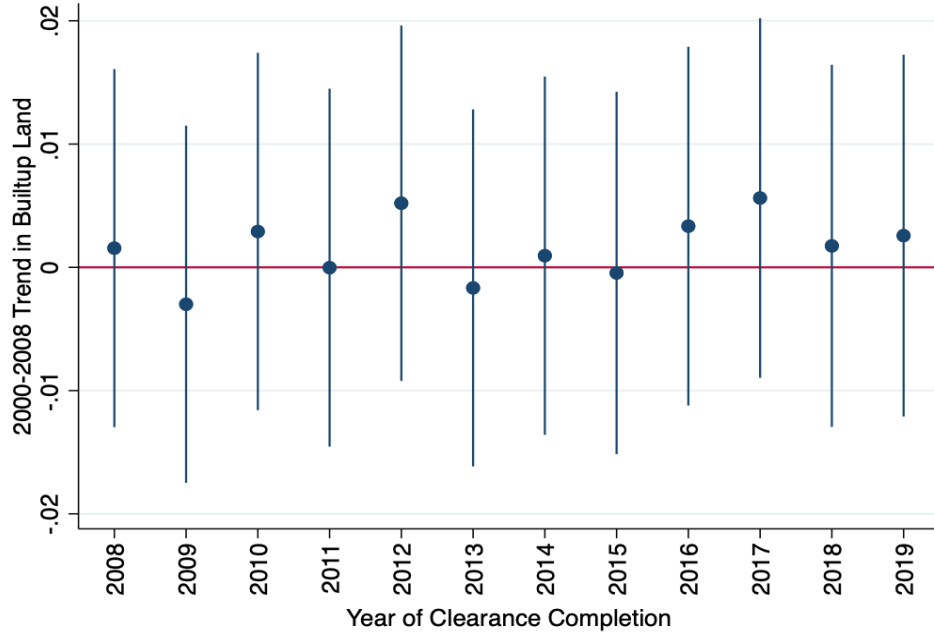
where $LandUse_{uipt}$ represents the share of the hazardous area polygon i in province p

classified as in use for land use u in year t . We conduct the analysis separately for each land use category u , so obtain specific parameters for each category. β_u is our parameter of interest, denoting the treatment effect associated with $Cleared_{jpt}$, a binary variable taking a value of 1 if hazardous area i is cleared of landmines at year t , and 0 otherwise. δ_{ui} is a vector of hazardous area FEs, which capture time-invariant local characteristics. δ_{upt} is a vector of province-by-year FEs, which capture geographically invariant characteristics within each year-by-province combination. We cluster standard errors by district and year.

Our land use analysis leverages the finer units of analysis of the specific hazardous area polygons, as well as the extended time window of our analysis sample, which now goes up to 2020. This allows us to capture both longer tailed impacts as well as those occurring in unelectrified rural areas. Moreover, our treatment measure now reflects the clearance of the full hazardous area observation (rather than the final clearance of a grid cell, where intervening clearance of some areas may have already occurred).

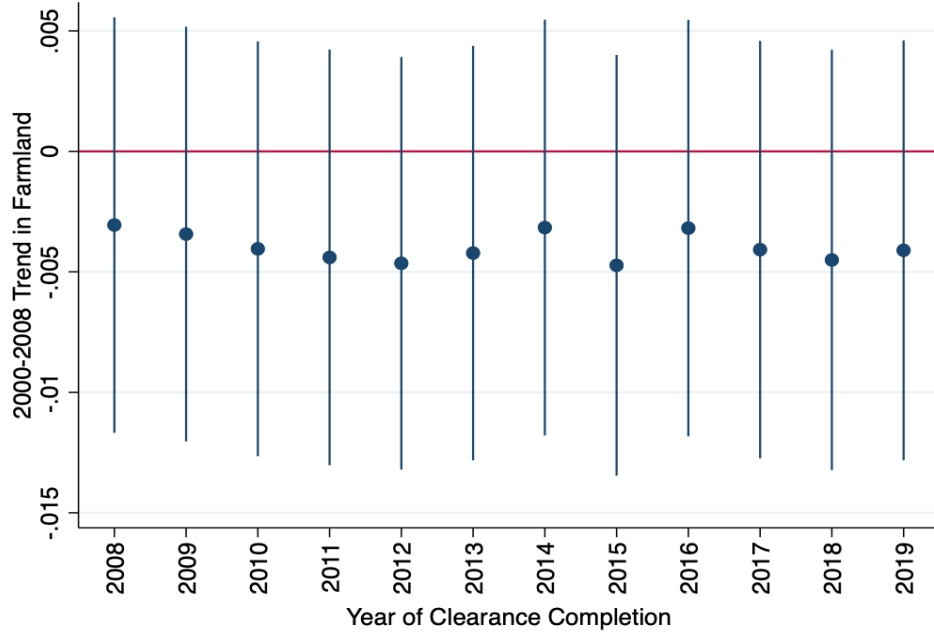
As with our NL analysis, our land use analysis is predicated on the parallel trends assumption. We thus also document that there are no differential changes in land use categories between 2000 and 2008 that correlate with the timing of each hazardous areas' clearance. Figure 6 shows the pre-trends in built-up land use, while Figure 7 shows the pre-trends in farmland land use.⁷

Figure 6: Pre-trends in Built-Up Land



⁷Pre-trends for additional land use classifications can be found in the appendix.

Figure 7: Pre-trends in Farmland



The main challenge of the land use analysis is that the computational demands of processing all of the Landsat imagery limit us to conducting our analysis at five year rather than annual intervals. As such, event study-based approaches such as those we adopt in our NL analysis are not feasible for our land use analysis. However, given the apparent robustness of our standard two-way FE model in our NL estimations, we expect our land use results to be similarly valid.

4.3 Conflict

We similarly leverage the roll-out of clearance activities to estimate the impact of clearance on subsequent conflict. Our data here are at more aggregate 10km grid cell scales, but our time window now stretches until 2020 at annual frequency.

The estimating equation for this analysis is:

$$Conflict_{kpt} = \alpha + \beta Cleared_{kpt} + \delta_k + \delta_{pt} + \epsilon_{kpt}$$

where $Conflict_{kpt}$ is a count of conflict events in grid-cell k in province p during year t . $Cleared_{kpt}$ is a binary variable taking a value of 1 if grid cell k is entirely cleared of landmines at year t , and 0 otherwise. δ_k is a vector of grid-cell FEs, which capture time-invariant local characteristics. δ_{pt} is a vector of province-by-year FEs, which capture geographically invariant characteristics within each year-by-province combination. As the area of each grid-cell intersecting with hazardous areas varies, this analysis is weighted by the percent of grid cell k covered by hazardous areas.

5 Results

5.1 Nighttime Lights

We begin by estimating the effect of landmine clearance on economic development among hazardous areas cleared during or after 2008, using NL. First we examine the event-study results using the [Chaisemartin and d’Haultfoeuille \(2020\)](#) dynamic estimator, shown in Figure 8. Confirming our primary assumption, we see little evidence of meaningful changes in NL prior to the full clearance of a grid cell. However, NL output seems to increase consistently with each additional year after clearance. These effects are quite large, with increases of more than the pre-treatment mean NL emissions after only 3 years, suggesting that clearance of hazardous areas can rapidly alter the trajectory of economic output and conditions.

Figure 8: Event Study of Clearance Impacts on NL
Dynamic Estimator ([Chaisemartin and d’Haultfoeuille, 2020](#))

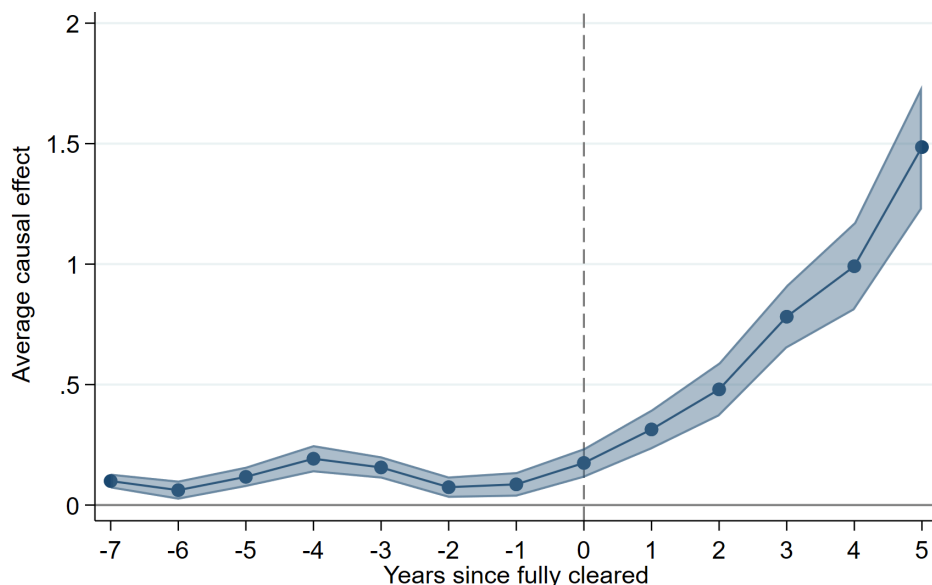
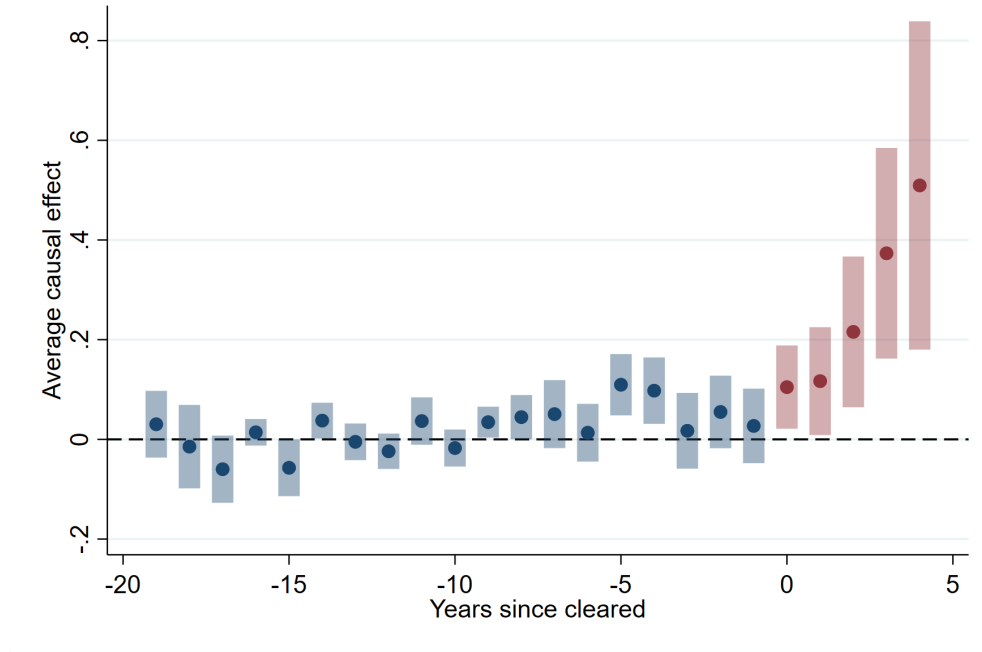


Figure 8 also indicates dynamic treatment effects that mount over time (rather than a one-time change in NL levels post-treatment that remains constant over time thereafter). This finding also validates our use of the dynamic estimator, as we would expect the treatment effects to be otherwise attenuated by comparisons of changes in treatment among late-treated cells with means among already-treated cells (whose NL levels would be continuing to rise).

Multiple approaches to addressing the potential bias from dynamic and heterogeneous treatment effects are now available. We thus also utilize the [Callaway and Sant’Anna \(2021\)](#) estimator, which makes a related but distinct set of identifying assumptions to recover unbiased event study results in the presence of dynamic treatment effects. The results, shown below in Figure 9, are nearly identical to our prior findings, with little evidence of upward trends in NL preceding clearance and mounting effects beginning shortly after clearance.

Figure 9: Event Study of Clearance Impacts on NL
Dynamic Estimator (Callaway and Sant’Anna, 2021)



Nonetheless, we also use (non-dynamic) two-way FE estimation to explore heterogeneity in the effects of clearance on NL for the sample cleared between 2008 and 2013. Table 3 first shows that we continue to find significant impacts of clearance on NL even in this specification (Column 1). We then examine how these effects vary by road access⁸ and population density.⁹ The impacts of clearance are strongest among the cells located closest to a road (Column 2), and diminish to approximately zero at more than 4.5 km away from a road. In Column 3, we also find that more populated areas experience larger NL impacts from clearance. While the effects are largest in areas with the greatest initial population levels, the estimates imply relatively large effects even in low population areas. For example, these estimates imply that areas with population densities of only 200 inhabitants/km² (classified as “rural” in most schema) see NL gains equal to 100% of the control mean (0.459), effectively doubling their NL due to clearance.

⁸We construct a *Distance to Roads* measure that reflects the distance from the grid cell to the nearest road on the United Nations Office for the Coordination of Humanitarian Affairs (OCHA) road network map.

⁹To reflect the baseline population density for the broader area in which each grid cell is located, we use the district mean population density in the year 2000 from the CIESIN Gridded Population of the World Version 4.

Table 3: Clearance Impacts on NL

DV=Nighttime Lights	(1)	(2)	(3)
Cleared of Landmines	.699*** (0.209)	1.047*** (0.296)	0.0801 (0.162)
Cleared * Dist. to Road		-0.230** (0.0825)	
Cleared * Baseline Pop.			0.00220*** (0.000538)
Observations	121,968	121,968	121,968
R-squared	0.627	0.629	0.646
Control Mean	0.459	0.459	0.459

*** p<0.01, ** p<0.05, * p<0.1. All models include grid cell and province-year fixed effects, and are weighted by percent cell covered by hazardous area. Standard errors clustered by district and year in parentheses.

Taken together, these NL impacts suggest important gains in economic conditions around a hazardous area once it is cleared. Since a hazardous area is at most $\sqrt{2}\text{km} \approx 1\text{mi}$ away from any point within the same grid cell, these effects suggest material economic changes even at reasonably small distances from cleared hazardous areas. Finding important nearby gains from clearance of hazards even in sparsely populated areas also contrasts with [Chiovelli, Michalopoulos and Papaioannou \(2018\)](#), who find primarily broader effects and ones primarily from clearing major roadways.

5.2 Land Use

We next consider the extent to which clearance of the hazardous areas altered the land use of these specific polygons using classes derived from MS imagery. In Table 4, we show impacts on built-up land use (reflected as the share of the polygon classified as built-up). We find significant increases in built-up land use after clearance (Column 1), equal to an increase of approximately 2 percentage point (over a sample mean of $\approx 11\%$). These are meaningful changes, given that construction of new built-up uses or expansion of existing ones can reflect larger investment in a context of challenging economic and security conditions.

Another advantage of the land use classification at this scale is that it enables us to unpack these effects by the types of blockages in each hazardous area. In Columns 2-7 of Table 4, we show separate estimations for sub-samples of polygon blockage types as designated by the clearance teams and recorded in the DMAC database (grazing, agriculture, road, housing infrastructure and water blockages). We find the gains are concentrated in polygons facing grazing and agriculture blockages, which comprise the vast majority of the sample. We find relatively few gains in built up use in polygons facing road, infrastructure, or housing blockages. At first glance, this may seem counter-intuitive, as grazing and agricultural blockages occur in areas that are not as built-up on average as other blockage types (such as housing and infrastructure). However, clearing landmine hazards from these areas could

allow the expansion of built-up uses to a greater degree than increasing density of built-up uses in already heavily built-up areas. The blockages initially recorded by the clearance teams were predictions, but clearing grazing and agricultural areas could have led to a broader set of uses that were not originally envisioned (in this case, particularly built-up uses). These effects appear particularly important in villages and small towns where grazing and agricultural blockages most frequently occur (and which comprise the vast majority of our sample).

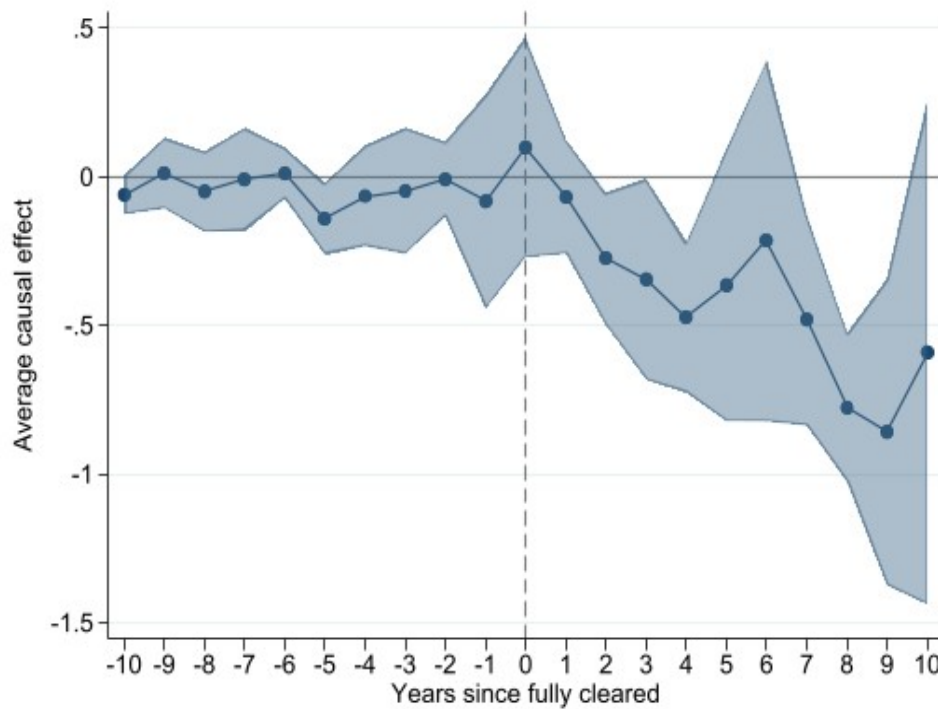
Table 4: Impacts on Built-up Land Use

DV = Pct. Built-Up	(1) All	(2) Grazing	(3) Agriculture	(4) Road	(5) Housing	(6) Infrastructure	(7) Water
Cleared	0.0227** (0.00728)	0.0212** (0.00756)	0.0326 (0.0171)	0.0114 (0.0122)	0.00983 (0.0202)	-0.00398 (0)	0.0794** (0.0293)
Observations	33,972	26,874	6,366	2,604	2,130	720	582
R-squared	0.656	0.664	0.641	0.649	0.684	0.702	0.740
Hazard FEs	Y	Y	Y	Y	Y	Y	Y
Year*Prov. FEs	Y	Y	Y	Y	Y	Y	Y
Pre-clearance buildup	.11	.104	.113	.101	.16	.308	.149

Column headings reflect blockage type subsamples
Standard errors in parentheses clustered by district and year
*** p<0.01, ** p<0.05, * p<0.1

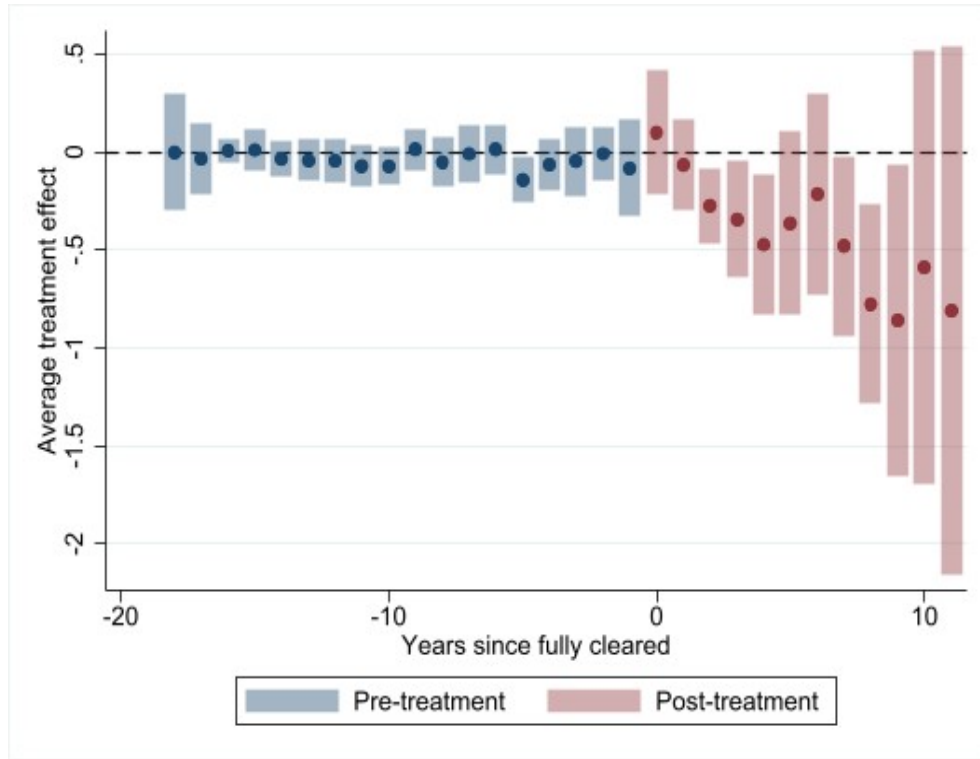
As a corollary to the increasing prevalence of built-up uses on newly cleared lands originally classified as blocked for grazing and agricultural uses, we investigate whether these areas exhibit increasing farmland uses. In Table 5, we show impacts of clearance on the share of each polygon classified as a farmland use. In Column 1, we find no material effects on the full sample of polygons. In Columns 2 and 3, we repeat this estimation for subsamples of polygons categorized as grazing and agricultural blockages, again finding no evidence of impacts of clearance on the reclassification of these areas into farmland uses. It is possible that the changes in farming on these lands were not dramatic enough to be observable as changes in the land use classification. For example, in some cases, farmers may have already attempted to sow and harvest wheat even in hazardous areas, and intensified their planting after clearance (and thus be classified as farmland both pre- and post-clearance). However, such examples are likely outliers, as landmine hazards posed major risks and many of the polygons designated as agricultural blockages are not classified as farmland uses at baseline. A more likely explanation may be that water availability and access to fertilizer and other inputs may be the binding constraints to farming expansion in this context.

Figure 10: Event Study of Clearance Impacts on Conflict
Dynamic Estimator ([Chaisemartin and d'Haultfoeuille, 2020](#))



These results are quite consistent with those for NL, exhibiting mounting improvements after clearance that grow with time. We also consider the alternative approach of [Callaway and Sant'Anna \(2021\)](#) in Figure 11, finding very similar results.

Figure 11: Event Study of Clearance Impacts on Conflict
Dynamic Estimator (Callaway and Sant’Anna, 2021)



We next consider how conflict intersects with NL and land use conditions. We first return to our NL grid cell sample and incorporate both contemporaneous and baseline conflict conditions in the 5km grid cell into which each 1km NL grid cell falls. We split our sample into high and low baseline conflict subsamples (i.e., above/below median baseline conflict), with results for the former in Table 6 and latter in Table 7. In Columns 1 of each respective table, we find quite similar estimates of clearance impacts on NL (if anything, effects are slightly larger in the “high baseline conflict” subsample). These suggest that landmine clearance can be an effective development strategy even in areas that have experienced substantial conflict risks.

We next investigate whether the clearance impacts on satellite-based outcomes are correlated with the impacts on conflict risks. This can be framed as asking whether the same locations experience impacts on both dimensions, or whether these improvements are independently distributed across locations. To do so, we include nearby contemporaneous conflict as a covariate (recognizing that these are themselves endogenous outcomes of the clearance measure). If the impacts on both dimensions are geographically clustered (i.e., the same locations largely experience both gains in NL and conflict reductions), we would expect the coefficient on clearance to be attenuated toward zero.

However, when we add contemporaneous conflict as a covariate in both models in Columns 2 of Tables 6 and 7, we see no material difference in the coefficient on landmine clearance from the base models. At the same time, we do find significant negative correlations between contemporaneous conflict and NL outcomes. Taken together, these suggest that the

clearance can also reduce the likelihood of further conflict in these areas. Both dimensions of improvements are important from a welfare perspective. Although it is possible that conflict reductions due to clearance could have knock-on effects on subsequent economic activity, we find no evidence of clustering of these outcomes, suggesting improvements in one dimension are not necessarily dependent on improvements in the other.

6 Conclusions

We offer the first evidence that clearing areas potentially contaminated with landmines can reshape economic activity even in the midst of ongoing conflict. The limited work on landmine clearance impacts to date has focused on post-conflict settings, finding the greatest gains in clearing transport routes linking major markets. Using spatially precise clearance data and satellite imagery, we find that clearing areas in rural settings can also spur new development. The largest changes appear to be the improvement and expansion of buildings and small-scale infrastructure in small towns and villages. These shifts are particularly pronounced in locations with the greatest conflict risks.

At the same time, landmine clearance also appears to reduce the risks of further conflict directly. This may be because landmine explosions can trigger subsequent violence in response, or because they weaken the broader sense of safety and rule of law in an area. Whatever the pathway, landmine clearance helps to avert further risks beyond those due to the landmine themselves.

Clearing a hazardous area of potential landmines remains a painstaking and dangerous task, and the sector remains underfunded given the geographic scope of areas still considered hazardous (even ignoring newly laid landmines). Although a full benefit-cost analysis is beyond the scope of our study, our results suggest there are important economic gains beyond those represented by the averted direct harm of the mines to the people they injure or kill. Our work also suggests that more equitably targeting rural and urban areas for landmine clearance is warranted. Given the aforementioned budget constraints, more urban areas and major transport corridors are often given precedence both because their clearance is believed to induce the largest gains and because they are often easier to reach. Our results suggest a rebalancing may be at least partly in order, as there are important gains from clearance even in small towns and villages.

Finally, our findings point to some important areas for future research. First, although we do not observe changes in whether an area is farmed, there may be finer changes in the intensity and productivity of farming activities that could be detectable with MS-based data. Second, as climate change induces new pressures on economic activity—particularly in conflict-affected settings—the gains from landmine clearance may be magnified or dampened. A further stream of research may therefore tackle the intersection of clearance impacts and climate conditions.

Power to the People: Evidence from a Randomized Field Experiment of a Community-Based Monitoring Project in Uganda

Martina Björkman* and Jakob Svensson#

Abstract: Strengthening the relationship of accountability between health service providers and citizens is by many people viewed as critical for improving access to and quality of health care. How this is to be achieved, and whether it works, however, remain open questions. The paper presents a randomized field experiment on increasing community-based monitoring. As communities began to more extensively monitor the provider, both the quality and quantity of health service provision improved. One year into the program, there are large increases in utilization, significant weight-for-age z-score gains of infants, and markedly lower deaths among children. The findings on staff behavior suggest that the improvements in quality and quantity of health service delivery resulted from an increased effort by the staff to serve the community. Overall, the results suggest that community monitoring can play an important role in improving service delivery when traditional top-down supervision is ineffective.

World Bank Policy Research Working Paper 4268, June 2007

The Policy Research Working Paper Series disseminates the findings of work in progress to encourage the exchange of ideas about development issues. An objective of the series is to get the findings out quickly, even if the presentations are less than fully polished. The papers carry the names of the authors and should be cited accordingly. The findings, interpretations, and conclusions expressed in this paper are entirely those of the authors. They do not necessarily represent the view of the World Bank, its Executive Directors, or the countries they represent. Policy Research Working Papers are available online at <http://econ.worldbank.org>.

This project is a collaborative exercise involving many people. Foremost, we are deeply indebted to Frances Nsonzi and Ritva Reinikka for their contributions at all stages of the project. We would also like to acknowledge the important contributions of Gibwa Kajubi, Abel Ojoo, Anthony Wasswa, James Kanyesigye, Carolyn Winter, Ivo Njosa, Omiat Omongin, Mary Bitekerezozo, and the field and data staff with whom we have worked over the years. We thank the Uganda Ministry of Health, Planning Division, the World Bank's Country Office in Uganda, and the Social Development Department, World Bank, for their cooperation. We are grateful for comments and suggestions by Paul Gertler, Esther Du.o, Abhijit Banerjee, and seminar and conference participants at LSE, Oxford, IGIER, MIT, World Bank, NTNU, Namur, CEPR/EUDN conference in Paris, and BREAD & CESifo conference in Venice. Finally, we wish to thank the Bank-Netherlands Partnership Program (BNPP), the World Bank Research Committee, the World Bank Africa Region division, and the Swedish International Development Agency, Department for Research Cooperation, for funding this research.

Björkman also thanks Jan Wallander.s and Tom Hedelius.Research Foundation for funding.

*IGIER, University of Bocconi, and CEPR. Email: mbjorkman@unibocconi.it.

#IIES, Stockholm University, NHH, and CEPR. Email: jakob.svensson@iies.su.se.

1 Introduction

Approximately 11 million children under five die each year. Almost half of these deaths occur in Sub-Saharan Africa where roughly one in five children dies before reaching the age of five. More than half of these children – nearly 6 million – will die of diseases that could easily have been prevented or treated if the children had had access to a small set of proven, inexpensive services.¹

Why are these services not provided? While there is no simple answer, a wealth of anecdotal, and recently more systematic, evidence shows that the provision of public services to poor people in developing countries is constrained by weak incentives of service providers – schools and health clinics are not open when supposed to; teachers and health workers are frequently absent from schools and clinics and, when present, spend a significant amount of time not serving the intended beneficiaries; equipment, even when fully functioning, is not used; drugs and vaccines are misused; and public funds are expropriated.²

The traditional approach to accountability in the public sector relies on external control. This is a top-down approach where someone in the institutional hierarchy is assigned to monitor, control and reward/punish agents further down in the hierarchy. The tacit assumption is that more and better enforcement of rules and regulations will strengthen providers' incentives to increase both the quantity and quality of service provision. But, in many poor countries, the institutions assigned to monitor the providers are typically weak and malfunctioning, and may themselves act under an incentive system providing little incentives to effectively monitor the providers. As a result, the relationship of accountability of provider-to-state is ineffective in many developing countries.³

¹See Lancet (2003) and UNICEF (2003). It is estimated that 2 million children under five die from diarrhea, which in most cases can be treated with simple oral rehydration therapy. Another 2 million children die from pneumonia, where once more there is sufficient evidence of effective treatment (antibiotics). Malaria kills one million children under five, most of whom could have been protected by preventive measures and treatment with anti-malarias. Globally, neonatal disorders account for the highest proportion of deaths of children – many of them could have been saved if mothers had had access to basic antenatal and delivery care. Approximately half a million children under five die from measles, for which there is a cheap and effective vaccine (Black et al., 2003; Jones et al., 2003).

²For anecdotal and case study evidence, see World Bank (2003). Chaudhury et al. (2006) provide systematic evidence on the rates of absenteeism based on surveys where enumerators made unannounced visits to primary schools and health clinics in seven developing countries. Averaging across countries, 35 percent of the health workers were absent. Banerjee et al. (2004) and Duflo and Hanna (2005) confirm these findings. On misappropriation of public funds and drugs, see Reinikka and Svensson (2004) and McPake et al. (1999).

³In addition, while well-functioning legal and financial systems can curtail obvious cases of mismanagement, they only partially constrain the discretionary powers of public sector managers and employees. The complexity of the tasks performed by a typical public sector unit and its informational advantage relative to the monitor (which typically relies on accounting data) make it nearly impossible to design legal and accounting measures to address all types of misuse and thus, to curtail less obvious cases of mismanagement (such as shirking, budget prioritization in favor of staff and political considerations). Finally, audit reports and legal procedures are often difficult for nonspecialists to interpret and will therefore go unnoticed unless the commissioning agency acts on them.

As a complementary strategy, it has therefore been argued that more effort must be placed on strengthening beneficiary control, i.e. strengthening providers' accountability to citizen-clients (see e.g., World Bank, 2003). However, despite the enthusiasm for such an approach, there is little credible evidence on the impact of policy interventions aimed at achieving it (Banerjee and He, 2003; Banerjee and Duflo, 2005). This paper attempts to provide some.

To examine whether beneficiary control works, we designed and conducted a randomized field experiment in 50 "communities" from nine districts in Uganda.⁴ In the experiment, or intervention, communities were provided with baseline information on the status of service delivery, both in absolute terms and relative to other providers and the government standard for health service delivery. Community members were also encouraged to develop a plan that identified areas of concerns in health provision, and how to address them within the existing resource envelope.

The intervention sought to relax two constraints typically faced by communities in monitoring providers: lack of access to reliable and structured information on the community's entitlements and the status of service delivery, and inadequate local organizational capacity. Access to reliable and structured information about current status of service delivery and entitlements is critical for citizens' ability to monitor service providers. Although people know whether their own child died or not, and whether the health workers did anything to help them, they typically do not have any information on aggregate outcomes, such as how many children in their community did not survive beyond the age of 5 or where citizens, on average, seek care. Provision of information on outcomes and performance improves citizens' ability to challenge abuses of the system, since reliable quantitative information is more difficult for service providers to brush aside as anecdotal, partial, or simply irrelevant. But information provision may not have any considerable impact unless there are members of the community who are willing to make use of the new information. Exerting accountability (monitoring providers) is subject to potentially large free-rider problems. Elite capture further complicates the process of holding providers accountable. By enhancing local organization capacity and encouraging the community to develop its own monitoring strategy, these constraints are sought to be relaxed.

The community-based monitoring project increased the quality and quantity of primary health care provision. One year into the program, we find a significant difference in the weight of infants (0.17 z-scores increase) and a markedly lower number of deaths among children under five (a 33 percent reduction in child deaths) in the treatment communities. Utilization (for general outpatient services) was 16 percent higher in the treatment compared to the control facilities. We also find significant differences in the number of deliveries at treatment facilities and the use of antenatal care and family planning. Treatment practices, as expressed both in perception-based responses by households and in more quantitative indicators (immunization of children, waiting

⁴A "community" is operationalized as the households (and villages) residing in the five-kilometer radius around the facility (see section 5 for details). Approximately 110,000 households (600,000 individuals) reside in these communities, of which half reside in the treatment communities.

time, examination procedures, absenteeism), improved significantly in the treatment communities, thus suggesting that the changes in quality and quantity of health care provision are due to behavioral changes of the staff. We find evidence that the treatment communities became more engaged and began to monitor the health unit more extensively. No effect is found on investments, or the level of financial or in-kind support (from the government). Furthermore, supervision of providers by upper-level government authorities remained low in both the treatment and the control group. This reinforces our confidence that the findings on the quality and quantity of health care provision resulted from increased efforts by the health unit staff to serve the community in the light of better community monitoring.

The paper is organized as follows. The next section reviews the literature. Section 3 discusses the concept of community monitoring. Section 4 briefly describes the institutional environment in Uganda and in the project areas. The community-based monitoring intervention is described in section 5. Section 6 lays out the evaluation design and the results are presented in section 7. Section 8 concludes.

2 Literature Review

Improving governance and public service delivery through community participation is an approach that has gained prominence in recent years. For example, the World Development Report 2004 is entirely devoted to the concept of increasing poor citizens' voice and participation in service delivery in order to help them monitor and discipline providers. But despite the enthusiasm for such approaches, there is little credible evidence on the impact of policy interventions aimed at achieving them. On the one hand, most (all) comprehensive community based monitoring initiatives have not been rigorously evaluated. On the other hand, the few studies relying on rigorous impact evaluation strategies have not evaluated more comprehensive attempts to inform and involve the community in monitoring public officials.

On the latter issue, Olken (2005) evaluates different ways of monitoring corruption in a road construction project in Indonesia. In one of the experiments, invitations were sent out to village-level meetings where project officials documented how they spent project funds for local road construction. However, although the invitations increased the number of people participating in the meetings, the meetings were still dominated by members of the village elite. Moreover, corruption is not easily observable and project officials may very well be able to hide it when reporting on how funds were used. The data also reveal that corruption problems were seldom discussed in these meetings.⁵ Thus, it is unclear to what extent non-elite community members were

⁵The information problem is illustrated in the novel but burdensome way in which Olken (2005) estimates the extent of corruption. Specifically, Olken (2005) assembled a team of engineers and surveyors who dug samples in roads to estimate the quantity of materials used and then, using price information from local supplies, estimated the extent of "missing" expenditures. The corruption estimates were not reported in the village meetings.

complaints or social opprobrium (Banerjee and Duflo, 2005). Similarly, they may be able to induce higher effort of health workers by providing non-pecuniary rewards (social rewards) for good performance. To the extent that the service is valuable to them, they should also have strong incentives to monitor the provider – incentives which the external agent assigned to supervise the provider may lack. Naturally, there is no guarantee that community monitoring will work even if the community is informed, can coordinate actions and there is demand for the service. In many developing countries, the beneficiaries of health services in rural areas are socially inferior to health care workers. Beneficiary groups may also be captured by the service provider or other authorities through their social or political connections (Banerjee and Duflo, 2005). Thus, in the end, if and to what extent community monitoring works is an empirical question.

4 Institutional Setting

Uganda, like many newly independent countries in Africa, had a functioning health care system in the early 1960s. Accessibility and affordability were relatively extensive. The 1970s and 1980s saw the collapse of government services as the country underwent political upheaval. Health indicators fell dramatically during this period until peace was restored in the late 1980s. Since then, the government has been implementing major infrastructure rehabilitation programs in the public health sector. Some health indicators have improved, while others have not. This is despite a GDP growth rate exceeding 64 percent and a 40-percent reduction in consumption poverty in the 1990s (Appleton 2001)

As of 2001, public health services are free of charge. Anecdotal and survey evidence (see below), however, suggests that users still encounter varying costs when visiting public health facilities.

The health sector in Uganda is composed of four types of facilities: hospitals, health centers, dispensaries (health center III), and aid posts or sub-dispensaries. These facilities can be government, private for-profit, or private not-for-profit operated and owned. The impact evaluation focuses on dispensaries. Dispensaries are in the lowest tier of the health system where a professional interaction between users and providers takes place. Most dispensaries are rural (89 percent). According to the government health sector strategic plan, the standard for dispensaries includes preventive, promotional, outpatient care, maternity, general ward, and laboratory services (Republic of Uganda 2000). In our sample of facilities, on average, a dispensary was staffed by an in-charge or clinical officer (a trained medical worker/doctor), three nurses (including midwives), and three nursing aids or other assistants.

The health sector in Uganda is decentralized and supervision and control of the dispensaries are governed at the district level. A number of actors are responsible for the functioning of the dispensaries. The Health Unit Management Committee (HUMC) is supposed to be the main link between the community and the health

facility. Each dispensary has an HUMC which consists of members from both the health facility staff and non-political representatives from the community (elected by the sub-county local council). The HUMC should monitor drugs and finances disbursed to the health facility, as well as the day-to-day running of the health facility (Republic of Uganda 2000). The HUMC can warn the health facility staff on issues of indiscipline, rudeness to patients and misappropriations of funds by recommending that the staff is transferred from the health facility. However, the HUMC has no authority to dismiss a worker. In cases of problems at the health facility, the working practice is that the chair person of the HUMC raises the issue with the in-charge. If there is no improvement, the issue should be referred to the Health Sub-district.

The Health Sub-district monitors funds, drugs and service delivery at the dispensary. Supervision meetings by the Health Sub-district are supposed to appear quarterly but, in practise, monitoring is infrequent. The Health Sub-district has the authority to reprimand, but not dismiss, health facility staff for indiscipline. In severe cases of indiscipline, therefore, the errand will be referred to the Chief Administrative Officer of the District and the District Service Commission, which is the appointing authority for the district and has the authority to suspend or dismiss staff.

Another actor in the health sector is Community-based organizations (CBOs). Their main focus is on health education in antenatal care, family planning, and HIV/AIDS prevention.

5 The Project: Citizen Report Card

In response to perceived continued weak health care delivery at the primary level, a pilot project (Citizen report cards) aimed at enhancing community involvement and monitoring in the delivery of primary health care was initiated in 2004. The project was designed by staff from Stockholm University and the World Bank, and implemented in cooperation with a number of Ugandan practitioners and 18 community-based organizations. The 50 project facilities (all in rural areas) were drawn from nine districts in Uganda (see the appendix for details).

With the catchment area (or the community) of each dispensary defined as the households and villages residing within a five-kilometer radius from the clinic, about 110,000 households reside in the communities supposedly served.⁸ The facilities were first stratified by location (districts) and then by size (the number of households residing in the catchment areas). From each group, half the units, with corresponding catchment areas, were randomly assigned to the treatment group and the remaining 25 units were assigned to the control group. Each district thus had both treatment and control groups.

The main objective of the Citizen report card project was to strengthen providers' accountability to citizen-clients by enhancing communities' ability to monitor providers.

⁸Dispensaries are designed to serve households in a catchment area roughly corresponding to the five-kilometer radius around the facility (Republic of Uganda, 2000).

Specifically, the project aimed at: (i) providing communities with baseline information on the status of service delivery, both in absolute terms and relative to other providers, and the government standard for health service delivery at the dispensary level; and (ii) encouraging people to develop a plan that identified areas of concerns in health provision, and how to address them within the existing resource envelope. These components are discussed next. A time-line and a schematic view of the intervention and expected outcomes are depicted in figures 1 and 2.

5.1 Data Collection and Report Cards

Data collection was governed by two objectives. First, data were required to assemble report cards on how the community at large views the quality and efficacy of service delivery. We also wanted to contrast the citizens' view with that of the health unit staff. Second, data were required to rigorously evaluate impact. To meet these objectives, two surveys were implemented: a survey of health care providers and a survey of health care users. Both surveys were implemented prior to the intervention (data from these pre-intervention surveys formed the basis for the report cards) and one year after the project had been initiated.

A quantitative service delivery survey (QSDS) was used to collect data from the health service providers. Since agents in the service delivery system may have a strong incentive to misreport (or not report) key data, the data were obtained directly from the records kept by facilities for their own need (i.e. daily patient registers, stock cards, etc.) rather than from administrative records submitted to the district-level government. The former, often available in a highly disaggregate format, were considered to suffer the least from any incentive problems in record-keeping.

The household survey collected data on both households' health outcomes and health facility performance, including performance parameters such as usage, availability, access, reliability, quality and satisfaction. To the extent that it was possible, household responses were supported by patient records, i.e., patient exercise books and immunization cards. These records helped the household recall details about its visits to the health facility and also minimized problems of misreporting. The post-intervention household survey also included a shorter module on health outcomes. Specifically, data on under-five mortality were collected and we measured the weight of all infants in the surveyed households.

A stratified random sample of households within the catchment area of the facility were surveyed. In total, roughly 5,000 households have been surveyed in each round. The design and implementation of the surveys are explained in more detail in the appendix.

The data from the two pre-intervention surveys were analyzed and a smaller subset of the findings were assembled in report cards for the treatment localities.⁹ The data

⁹Thus, the design and size of the surveys were largely driven by the second objective – to evaluate impact.

included in the report cards were identified as key areas subject to improvement and include utilization, quality of services, informal user charges and comparisons vis-à-vis other health facilities in the district and the country at large. Each treatment facility and its community had a unique report card summarizing, in a format easily accessible to the communities, the findings from the surveys conducted in their area.

The report cards were translated into the main language spoken in the community.¹⁰ To support the non-literate community members, posters were designed by a local artist so that otherwise complex information and concepts were easily understood. Because the information in the report cards was largely statistical, the posters visually conveyed the main messages, such as where people go to seek medical care and why they do so.¹¹

5.2 Dissemination and Participation

Getting people to retain and use information to achieve a specific objective is a complex problem.¹² Extensive piloting concluded that simply reporting the facts would be likely to have little impact. Thus, to maximize the likelihood that the information in the report cards would be used when people decide what actions to take, a participatory approach was chosen where community members themselves actively interpreted and analyzed the information.¹³ To this end, the process of providing information and encouraging participation and monitoring was initiated through a series of meetings: a community meeting; a staff meeting; and an interface meeting. Staff from various Community-based organizations (CBO) acted as facilitators in these meetings.¹⁴

The community meeting was a two-day (afternoons) event with approximately 100 invited participants drawn from the surveyed villages in the catchment area of the health facility. To avoid elite capture, the invited participants consisted of a selection of representatives from different spectra of society (i.e. young, old, disabled, women,

¹⁰In the end, the report cards were translated into six different languages: Ateso (Soroti), Lusoga (Iganga), Lango (Apac), Luganda (Masaka, Wakiso, Mukono and Mpigi), Runyankore (Mbarara) and Lugbara (Arua).

¹¹See the appendix for a prototype poster.

¹²See, for example, Lupia (2004) who systematizes and draws conclusions from clinical, psychological, and economic research on information transmission and processing.

¹³The approach we used is closed linked to the so-called "participatory rural appraisal approach". Participatory rural appraisal (PRA) is a label given to a growing family of participatory approaches and methods with the common aim of enabling people to make their own appraisal, analyses, and plans. PRA evolved from a set of informal techniques used by development practitioners in rural areas to collect and analyze data (World Bank, 1996).

¹⁴Since the CBOs were in regular interaction with the communities and had a mandate drawn from a long-term presence on the ground working with the community, these facilitators were perceived to be a good conduit through which the project could be delivered. The CBO facilitators were trained for seven days in data interpretation and dissemination, utilisation of the participatory methodology, and conflict resolution and management. It should be noted that various CBOs (including some participating in the project) also operate in the control districts. Thus, the presence (and numbers) of CBOs in the project communities is similar across treatment and control groups.

mothers, leaders). The facilitators mobilized the village members by cooperating with village council representatives in the catchment area. Invited participants were asked to spread the word about the meeting and, in the end, a large number of uninvited participants from other villages who had found out about the event also attended the meeting. A typical village meeting was attended by more than 150 participants per day.

In the community meeting, the facilitators used a variety of methods, including maps, venn diagrams, role-play, and focus group discussions, to disseminate the information in the report cards in a participatory, or interactive, way.¹⁵ Information on patients' rights and entitlements was also discussed.¹⁶ As the objective was not only to inform but to encourage people to participate in developing a shared view on how to improve service delivery and monitor the provider, the facilitators structured the discussions through a series of questions on the various elements of accountability in the primary health sector (*who* is accountable to *whom*?; *what* is a particular actor accountable *for*?; *how* can these actors account for their *actions*?; and how are these elements reflected in the report card findings?). At the end of the meeting, the community's suggestions for improvements (and how to reach them without additional resources) were summarized in an action plan. The action plan contained information on health issues/services that had been identified as the most important to address; how these issues could be addressed and how the community could monitor improvements (or lack thereof). An abbreviated version of one such action plan is depicted in the appendix. While the issues raised in the action plans differed across communities, a common set of concerns included high rates of absenteeism, long waiting-time, weak attention of health staff, and differential treatment. After the meeting, participants were given posters and copies of the report card to bring back to their villages and share with their village members.

The health facility staff meeting was a one-day (afternoon) meeting held at the health facility with all staff present. In this meeting, the facilitators contrasted the information on service provision as reported by the provider with the findings from the household survey. The meeting enabled the providers to review and analyze their performance, and compare their performance with other health clinics in the district and across the country.

An interface meeting with participants (chosen by people that attended the community meeting) from villages in the catchment area and the health facility staff followed the community and health facility meetings. Based on the action plan developed in the community meeting and the discussions from the health facility meeting, the interface

¹⁵See the appendix for a more detailed description of the various methods used during the meetings.

¹⁶Information on patients' rights and entitlements was based on the Yellow Star program. In 2000, the MoH developed a quality of care strategy called the Yellow Star Program with the aim of improving and maintaining basic standards of care at government and NGO health facilities. The rationale behind this strategy was the general consensus that the quality of health services had been a major deterrent to service utilization. The Yellow Star Program lists a set of basic standards of quality. The standards fall into six categories: Infrastructure and Equipment; Management systems; Infection prevention; Information: Education and Communication; Clinical skills; and Client services.

meeting devised a strategy for improved health care provision. During the interface meeting, the community representatives and the health facility staff presented and discussed their suggestions for improvements. A role-playing exercise was used to disseminate the results from the survey, with community participants and staff reversing roles. The participants discussed their rights and responsibilities as patients or medical staff. The outcome was a shared action plan, or a contract, outlining the community's and the service provider's agreement on what needs to be done, how, when and by whom. The "community contract" also identified how the community could monitor the agreements and a time plan. Because the problems raised in the community meetings constituted the core issues discussed during the interface meetings, the community contract was in many respects similar to the community's action plan. Copies of the community contract were kept with the community and the health facility to support the following monitoring process.

5.3 Ongoing process of monitoring

The three separate meetings aimed at kick-starting the process of community monitoring. Thus, after the initial meetings, and based on the agreements in the community contract, the communities were themselves in-charge of establishing ways of monitoring the provider. The facilitators supported the communities in this process with follow-up meetings. This was done as an integrated part of the CBO's ordinary work in the villages. Each community had approximately two follow-up meetings in the six-month period that followed. In these meetings, facilitators raised the issues identified in the community contract with citizens and community leaders.

After a period of six months, the communities and health facilities were revisited to conduct a mid-term review – a repeat engagement on a smaller scale. Including a one-day community meeting and a one-day interface meeting, the review tracked the implementation of the community contract. The earlier community contracts were printed on posters to spark discussions. Health facility staff and community members jointly discussed suggestions on actions for sustaining or improving progress, or in the case of no improvements, why so. Where improvements had been made, suggestions for sustainability were recorded. The community and the health facility kept the updated community contract to assist in further monitoring.

6 Evaluation Design and Expected Outcomes

6.1 A stylized framework

The following sub-section presents a stylized framework that illustrates the channel(s) through which community-based monitoring could affect health outcomes.

As a robustness test, we also assess alternative explanations. Some of these alternative mechanisms are illustrated in figure 3. One concern is spillovers. Spillovers could affect the estimates in two ways. If information about the intervention spread to control areas and, as a result, control communities became more involved in monitoring the providers, the estimated treatment effect would be biased downward. If, on the other hand, households in control communities shifted from seeking care at the control facility to the nearest treatment clinic, it is possible that the estimated treatment effect would be biased upward. This is a potentially serious concern but also a mechanism which we can test. It is also possible that the intervention did not only (or primarily) increase the extent of community monitoring, but had an impact on other agents in the service delivery chain. For example, the various upper-level authorities in the health sector (e.g. the Health Sub-district) may have become more involved in monitoring the providers, or the district government may have increased its administrative or financial support, following the intervention. While this would not invalidate the causal effect of the intervention it would, of course, affect interpretation. Therefore, this alternative hypothesis is also subject to a battery of tests.

Given the wealth of information, we report the main results and tables in the text and refer the reader to appendix for additional findings.

6.3 Statistical framework

Given the randomized assignment of the Citizen Report Card project, we expect the 2004 pre-data in the treatment areas to be similar those in the control areas. We have both facility-specific data (on utilization, for example) and household-specific data (on waiting time, for example). Denoting y_{ijdt} the outcome variable of household i (when applicable), health facility j in district d and period t , we start by checking that there is no difference between treatment and control facilities/communities prior to the intervention:

$$y_{ijdtPRE} = \alpha_{PRE} + \beta_{PRE}T_{jd} + \varepsilon_{ijdtPRE} , \quad (1)$$

where $t = PRE$ denotes the pre-intervention period, T_{jd} is a dummy indicating whether health facility j is in the treatment group and $\varepsilon_{ijdtPRE}$ is the error term. In regressions using household data, the disturbance term is adjusted to allow for correlations within catchment areas (communities).

To estimate the causal effect of the program, we then run the same regression in the post-period ($t = POST$):

$$y_{ijdtPOST} = \alpha_{POST} + \beta_{POST}T_{jd} + \varepsilon_{ijdtPOST} . \quad (2)$$

We also estimate an extended version of equation (2):

$$y_{ijdtPOST} = \alpha'_{POST} + \beta'_{POST}T_{jd} + X_{ijdtPOST}\pi + \theta_d + \varepsilon_{ijdtPOST} . \quad (3)$$

As reported in table 2 (regressions 1-2), one year into the project, treatment facilities are significantly more likely to have suggestion boxes (no control facility had these, while 36 % of the treatment facilities did) and numbered waiting cards (only one control facility had these, while 25 % of the treatment facilities did). A higher share of treatment facilities also post information on free-services and patient's rights and obligations (regressions 3-4). The enumerators could visually confirm that 70 percent (17 out of 25) of the treatment facilities had at least one of these "monitoring tools" (suggestion boxes, numbered waiting cards, and/or posters on free-services), while only 4 out of 25 control units had at least one of them. The difference is statistically highly significant (column 5).

The results based on household data mirror the findings reported in columns 1-5. For example, the performance of the staff is more often discussed in local council meetings in the treatment communities (regression 6), suggesting that the treatment communities became more engaged. Three out of four households surveyed have attended at least one village meeting in 2005. Of those attending, 40 percent (13 percentage points) more households in the treatment community report that the functioning of the health facility was discussed. Combining the evidence from the CBO reports and the household survey data thus suggests that both the "quantity" of discussions about the project facility and the subject (from general to specific discussions about the community contract) changed in response to the intervention.¹⁹

7.3 Treatment practices

The qualitative evidence from the CBOs and, to the extent that we can measure it, the findings reported in table 2, confirm that the treatment communities became more involved in monitoring the provider. Did community monitoring affect the health worker's behavior and performance? We turn to this next. We report the results on treatment practices and staff behavior, both as expressed in perception responses by households (in appendix A.3) and in quantitative indicators such as the immunization of children, waiting time, staff absenteeism, examination procedures, management of the clinic, and extent of preventive care.

We start by looking at examination procedures.²⁰ Regression 1, table 3, presents the result of estimating (4) with the dependent variable being an indicator of whether any equipment (for instance thermometer or blood pressure equipment) was used during the examination. 49 percent of the patients in the treatment community reported that equipment was used the last time the respondent (or the respondent's child)

¹⁹ Additional evidence on community engagement and monitoring is reported in appendix A.3.

²⁰ Naturally, the relevant treatment is conditional on illness and the condition of the patient. However, since the project was randomly allocated across communities, there is no reason to believe that the type of illness and the condition of the patients should differ systematically across groups. In fact, we have information on reported symptoms for which the patient seeks care (from the household survey). There are, on average, no systematic differences in reported symptoms across treatment and control communities.

7.4 Utilization

The evidence presented so far shows that treatment communities began to monitor the health unit more extensively in response to the intervention and that in light of better community monitoring, the health unit staff responded by improving the provision of health services. We now turn to the question of whether increased community monitoring also resulted in improved quantity and quality of care.

Tables 7 and 8 report estimates of the treatment effect on quantity. We collected detailed data from the health facilities on the number of out-patients, the number of deliveries, the number of antenatal care patients, and the number of people seeking family planning services.²⁶

Table 7 presents the results, for the four different utilization variables, from the estimations of equations (2) and (4). There are positive and significant differences between treatment and control facilities across all four services. One year into the program, utilization (for general outpatient services) is 16 percent higher in the treatment facilities. When controlling for district fixed effects, the point estimate is slightly larger and more precisely estimated (significant at the 1 percent level). The difference in the number of deliveries at the facility (albeit starting from a low level) is even larger (68 percent, regression 4) and fairly precisely estimated. There are also positive and significant differences in the number of patients seeking antenatal care (22 percent, regression 8) and family planning (60 percent, regression 10).

As a complement to the difference approach, columns 3 and 6 present the results from the estimation of a value added specification.²⁷ Difference-in-differences estimates, i.e., equation (4), are reported in table A.10 in the appendix.²⁸ The point estimates from both specifications are positive and highly significant. The difference-in-differences estimates also suggest a larger treatment effect (28%) for outpatient services.

Table 8 reports changes in utilization patterns based on household data. We collected data on where each household member sought care during the last year in case of illness that required treatment. Apart from recording visits to the project facility (treatment or control facility), we recorded visits to private providers (for-profit and NGOs), traditional healers, self-treatment (i.e., purchases of medicine in drug shops), or other government facilities (i.e., not a project facility). Consistent with the findings reported in table 7, we find a positive and significant difference in the use of the project facility between treatment and control facilities (regression 1). The increase,

²⁶As discussed in section 5, these data were assembled by counting the number of patients from daily patient records, maternity unit records, the antenatal care register, and the family planning register.

²⁷Data on the number of antenatal care patients and the number of people seeking family planning services were not collected from medical records in the pre-treatment survey.

²⁸The value added specification is

$$y_{jdPOST} = \alpha_{VA} + \beta_{VA}T_{jd} + \lambda y_{jdPRE} + \varepsilon_{jdPOST} .$$

8.5 percent of the infants (up to 18 months) are severely underweight (< -3 z scores). Almost a quarter of the infants falls below the mildly underweight threshold (< -1 z score).

The difference in means of z scores of infants between the treatment and the control group is reported in regression 1, table 10. The estimated effect (difference) is 0.164 z score in weight-for-age. Regression 2 applies a more stringent restriction on the data to avoid problems of misreporting.³⁶ The difference in mean is 0.17 z score and is precisely estimated. Figure 6 plots the distribution of z scores for treatment and control groups. The difference in measured weight is most apparent for underweight children. Underweight status causes a decrease in immune and non-immune host defenses. Thus, since underweight children are at a higher risk of suffering from infectious diseases (and more severe complications of infectious diseases), and therefore in higher demand for/need of health care, the data in figure 6 are consistent with a positive treatment effect arising from improved access and quality of health care, rather than a general increase in nutritional status.

Regression 3 adds district fixed effects and controls for age and gender. The results remain qualitatively unchanged. The incidence of underweight increases with age. We cannot reject the hypothesis that the treatment effect is the same for girls and boys.

The treatment effect is quantitatively important. For this purpose, the baseline proportion of infants in each risk category (severe, < -3 z scores; moderately, $-3 \leq z$ scores < -2 ; mild, $-2 \leq z$ scores < -1) in the control group was calculated. Applying the shift in the weight-for-age distribution (adding 0.17 z score) with the odds ratio for each category – children who are mildly [moderately] {severely} underweight have about a two-fold [five-fold] {eight-fold} higher risk of deaths from infectious disease (Jones et al, 2003) – the reduction in average risk of mortality is estimated to be approximately 8 percent (figure 6).³⁷

Columns 4-5 in table 10 report the program impact on child weight for children between 18-36 months of age. The treatment effect is small and insignificant.³⁸

³⁶Specifically, we drop observations with a recorded weight above the 90th percentile in the growth chart reported in Cortinovis et al (1997). Since weight is measured by trained enumerators, the reporting error is likely due to misreported age of the child.

³⁷To put this into perspective, a review of controlled trials designed to improve the intake of complementary food for children aged six months to five years showed a mean increase of 0.35 z score (Jones et al, 2003). If the present coverage level were increased to universal coverage (99%), Jones et al estimate that complementary feeding alone would prevent 6% of the under-five deaths in the 42 countries with the 90% of worldwide child deaths in 2000. According to Jones et al, this is one of the most effective (in the sense of preventing under-five deaths) preventive interventions feasible for delivery at high coverage in a low-income setting.

³⁸Measurement errors due to misreported age of the child are likely to be a more serious concern for children above 18 months than for infants.

However, this does not seem to be the case either. The supervision of providers by upper-level government authorities remained low in both the treatment and the control group (table 16, regressions 1-2).

The incidence of supervision and control visits may be an imprecise measure of the effectiveness of monitoring by the upper-level authorities. A complementary measure is implemented sanctions. We have data on the extent to which staff was dismissed or transferred during the first year of the project. As noted in section 4, only the District Service Commission has the authority to dismiss and transfer staff. There is only a handful of staff that has been dismissed or transferred in 2005 and there is no systematic pattern that distinguishes treatment from control facilities (table 16, regressions 3-4). Likewise, there is no difference between treatment and control facilities in the number of staff that voluntarily left the facility during 2005 (regression 5).

Taken together, these findings reinforce our confidence that the improved quality and quantity of health care provision resulted from increased efforts by the health unit staff to serve the community in light of better community monitoring.

8 Conclusion

In this paper, we have studied the effects of enhancing rural communities' ability to hold primary health care providers accountable. We find that both the quality and quantity of the health service provision improved in the treatment communities: One year into the program, average utilization was 16 percent higher in the treatment communities; the weight of infants higher, and the number of deaths among children under five markedly lower. Treatment communities became more extensively involved in monitoring the providers following the intervention and the results suggest that the health unit staff responded by exerting a higher effort in serving the community. By strengthening the providers' incentives to serve the poor, health provision and in the end health outcomes can be significantly improved.

The starting point of this work is the mounting evidence showing that the provision of public services to poor people in developing countries is constrained by weak incentives of service providers. As argued in Chaudhury et al. (2006), this evidence is symptomatic of failures in "street-level" institutions and governance. However, although these failures constitute a direct obstacle to economic and social development, they have, until recently, received much less attention in the literature than weaknesses in macro institutions. This paper is an attempt at partly closing this gap.

Although the Citizen report card project appears to be successful, it is too early to use these findings as a basis for continued or increased support and funding for various activities with the aim of strengthening beneficiary control. There are still a number of outstanding issues. One important concern is to what extent the processes initiated by the Citizen report card project are permanent. Since the project is ongoing and scaled up to involve an additional 25 project units, this question can be answered at a